GEFORGE DESKTOP · PRODUCT BOOK

GeoForge Desktop User Manual

Make scientific models *actually run.*

From a scientific question to a reproducible model run. This book follows the interface you actually use and explains what happens, what the agent handles, and which decisions still belong to you.

ONE REPRODUCIBLE RUN 01—05

01 Describe your goal ↓

02 Agent selects a KI ↓

03 Prepare and validate data ↓

04 Run the real model ↓

05 Inspect plots, metrics, and files

Contents

00 Three ideas to remember

✦

Chat is primary

Describe the scientific task in normal language. GeoForge turns it into a visible, reviewable workflow.

KI

A KI is executable guidance

A KI tells the agent how to install, prepare, run, check, and interpret one scientific model.

⌂

Every project stays local

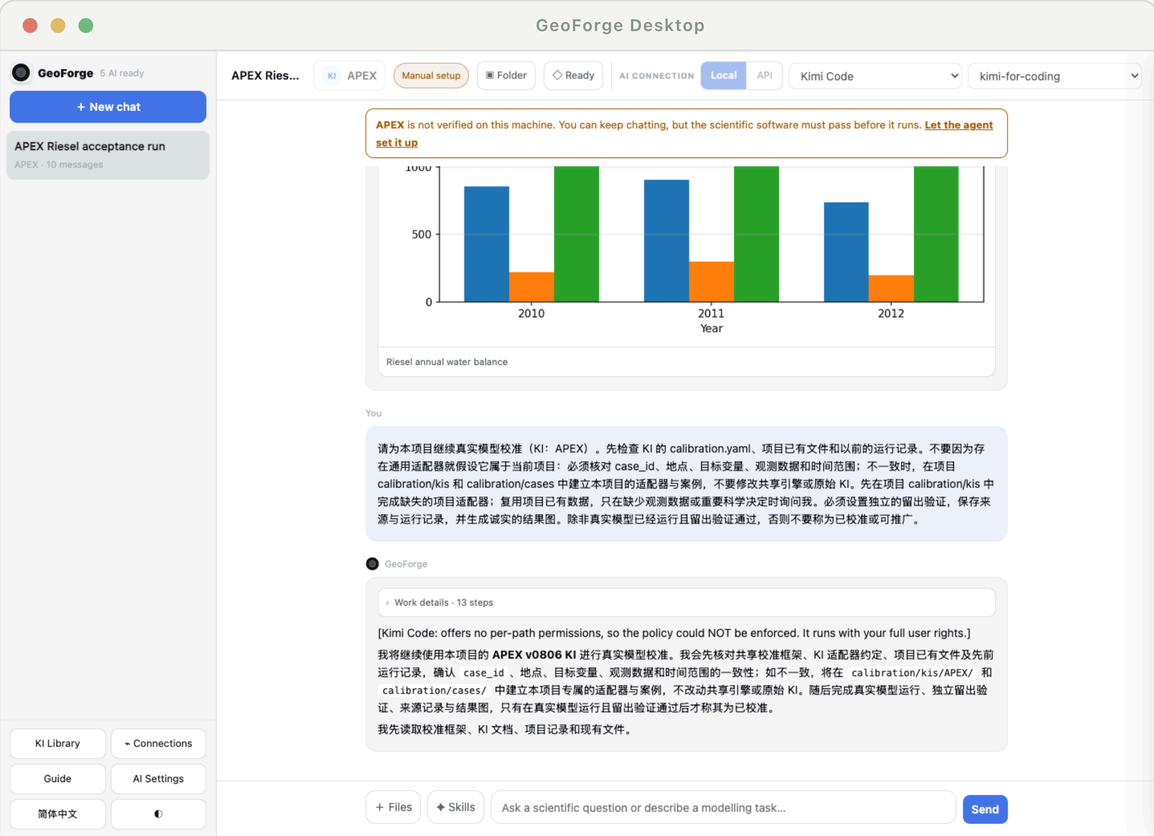
Messages, source data, model inputs, outputs, plots, and provenance live together in one folder on your Mac.

01

A 10-MINUTE PATH THROUGH GEOFORGE

Your first successful run

The shortest route is: create a local project, describe a concrete scenario, let the agent prepare what it safely can, answer only the decisions that truly need you, and verify the saved result.



PRODUCT SCREENSHOT

The chat stays central while project status remains visible at the side.

01.01

Five actions

1. Choose New conversation. Confirm the suggested project folder or select another writable folder.
2. Write one concrete request: place, period, process, and desired result. Missing details can be decided later.
3. Confirm the KI proposed by the agent. A KI may be available before its scientific software is verified on this Mac.
4. Open Project data. Use Resolve with agent for public or derivable data; add private files yourself when needed.
5. Let the agent run the real model. Review plots and metrics, then open the project folder to inspect the actual files.

01.02

How to know the run is real

- The project shows a completed model-run stage, not only a written plan.
- The outputs folder contains files created by the scientific model.
- The result cites the executable, input files, dates, parameters, and validation checks used.
- Plots are linked to saved data rather than being unsupported illustrations.

TIP A polished answer is not proof of a run. Saved inputs, command provenance, outputs, and checks are the proof.

02

FOLLOW ONE REAL PROJECT FROM A SENTENCE TO SAVED RESULTS

Worked example: run APEX for a 50 ha rainfed corn field

This chapter is the practical spine of the manual. Follow one real acceptance case near Riesel, Texas, for 2010–2012. You will create a local project, choose the APEX KI, let the agent prepare public weather and model inputs, respond to one data decision, run the verified APEX0806 executable, inspect the plots, and confirm the saved files. The screenshots are close-ups from the real GeoForge project rather than illustrations.

02.01

Know the target before clicking

The target is a project-contained acceptance run, not a publication-ready calibration study. The main scenario is a synthetic 50 ha rainfed corn field near Riesel. Public NASA POWER weather is used for 2010–2012, with a 20-year spin-up allowed by the KI. A successful run must leave real APEX outputs, validation metrics, plots, and provenance in the project folder.

- ✓ Model: APEX0806 through its documented Wine runtime on macOS.
- ✓ Inputs: weather, site, soil assumptions, crop management, and control files prepared by KI tools.
- ✓ Evidence: a completed real run, finite yields, a plausible water balance, plots, and provenance.

SUCCESS CHECK

You can restate the place, period, crop, area, and expected evidence in one sentence.

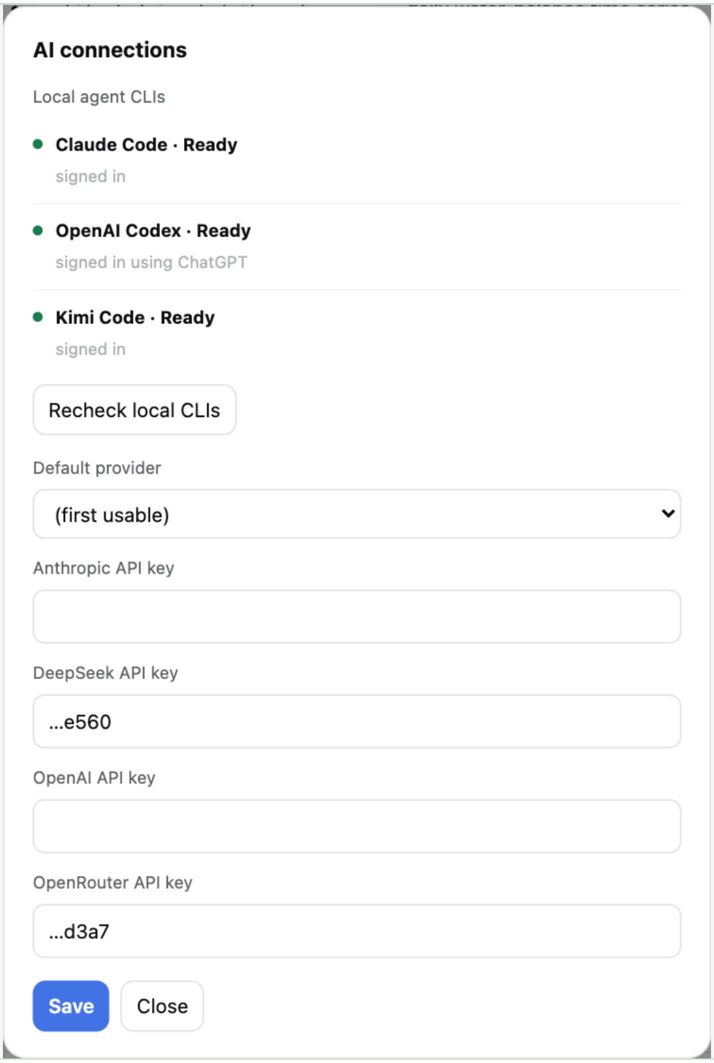
IF IT DOES NOT WORK

If your research question is still broad, begin with one site and a short period. Expand the case only after this first run succeeds.

02.02

Check that one AI connection is ready

1. Open AI Settings from the lower-left sidebar.
2. Choose either a ready local CLI (Claude Code, Codex, or Kimi) or a configured API such as DeepSeek.
3. For a local CLI, use Recheck after signing in through Terminal. For an API, save the key and run the connection check.



WHERE TO LOOK Read each provider's state here. Installed, signed in, and usable are separate checks.

SUCCESS CHECK

The provider is selectable in the chat header and no sign-in or update warning remains.

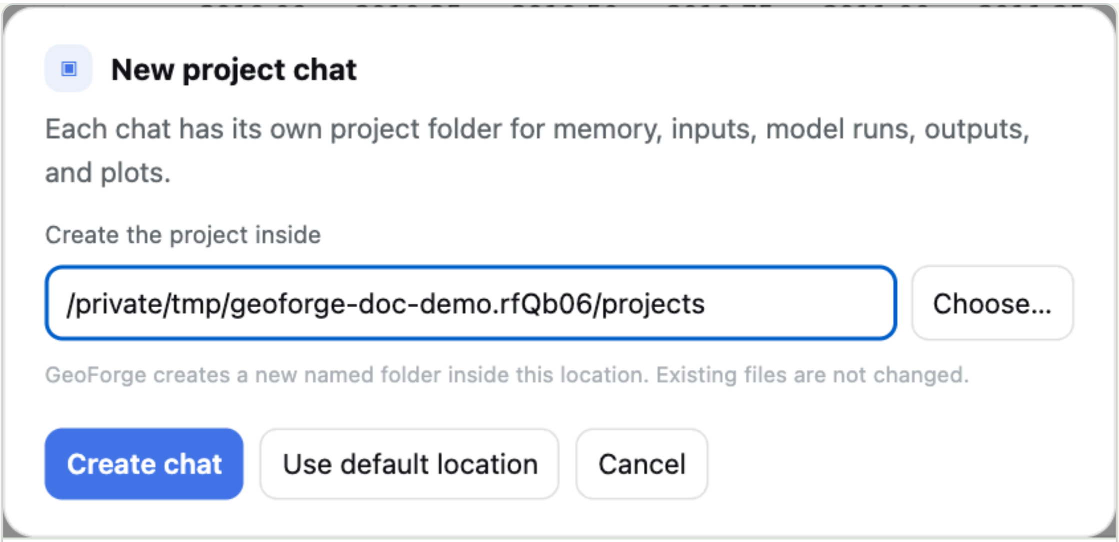
IF IT DOES NOT WORK

Open Terminal, run the provider's normal login command, return to GeoForge, and choose Recheck. Do not reinstall the scientific model to fix an AI login problem.

02.03

Create a project folder for this case

1. Choose New chat. GeoForge opens the New project chat window.
2. Use the suggested parent folder, or choose another writable local location.
3. Choose Create chat. GeoForge creates a new named subfolder; it does not modify existing files in the parent folder.



WHERE TO LOOK The path is visible before the conversation begins. This folder becomes the durable memory and model workspace.

SUCCESS CHECK

A new conversation appears in the sidebar and the path shown at the bottom of the window points to its own folder.

IF IT DOES NOT WORK

Choose a folder under your user Documents directory. Avoid a read-only disk, the application bundle, or a cloud folder currently syncing thousands of files.

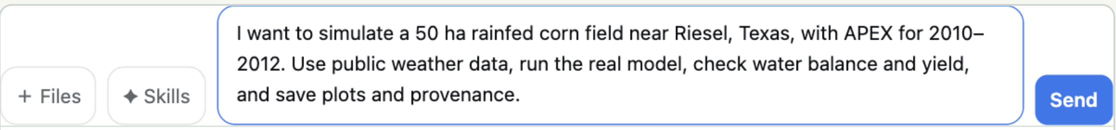
02.04

Describe the case in ordinary language

You do not need to list every APEX field. Give the agent the scientific scope and the evidence you want. The KI supplies the technical contract.

TYPE THIS REQUEST

```
I want to simulate a 50 ha rainfed corn field near Riesel, Texas, with APEX for 2010–2012. Use public weather data, run the real model, check water balance and yield, and save plots and provenance.
```



WHERE TO LOOK A concrete place, period, process, and output are enough to begin. The agent can ask for a real scientific decision later.

SUCCESS CHECK

The first agent response restates the scenario and says which KI it will use before running tools.

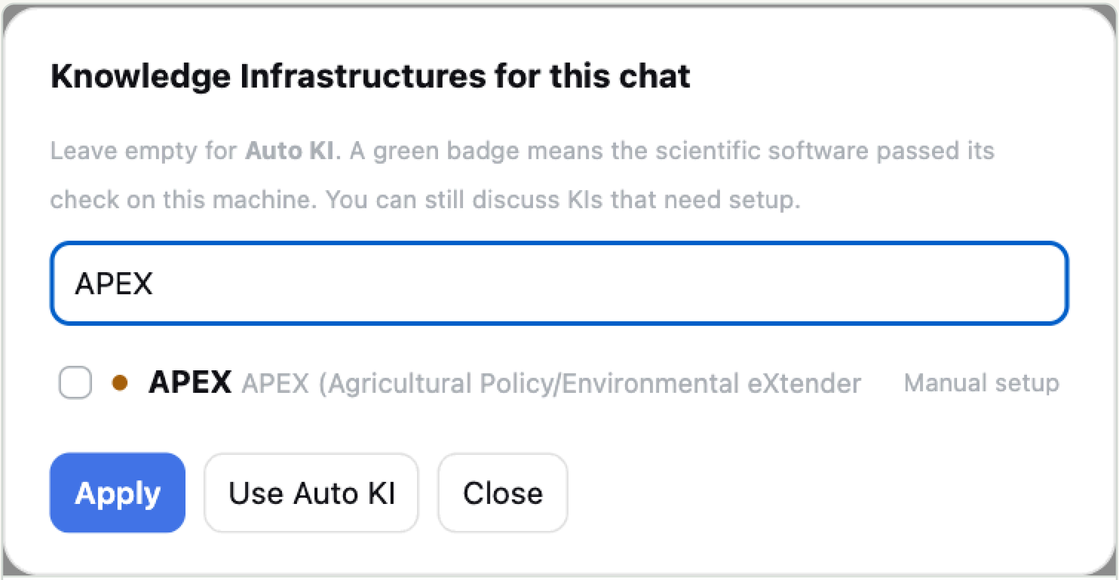
IF IT DOES NOT WORK

If the response is generic, add the missing place, dates, crop or process, and the result you expect. Do not paste dozens of unexplained model parameters.

02.05

Confirm the APEX KI

1. Choose the KI button in the chat header.
2. Search for APEX and select it. Leave Auto KI only when you genuinely want GeoForge to compare models.
3. Choose Apply. The selected KI is now fixed to this chat once the conversation starts running.



WHERE TO LOOK The badge reports whether the software has passed its local check; it is different from whether this project's data are ready.

SUCCESS CHECK

APEX appears next to the KI mark in the chat header.

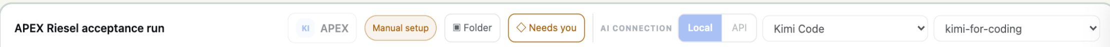
IF IT DOES NOT WORK

Open KI Library. If APEX is present but not verified, run its setup agent. If APEX is absent, import the KI package before returning to the chat.

02.06

Read the header before the run

- ✓ APEX tells you which KI is attached to this conversation.
- ✓ The verification badge refers to the scientific software on this Mac.
- ✓ Folder opens the current project's real directory in Finder.
- ✓ Project status opens the dynamic requirements, progress, data sources, calibration, and results panel.
- ✓ The right side shows the active provider and model for this chat.



WHERE TO LOOK These controls answer four different questions: which KI, whether software works, where files live, and what the project is doing.

SUCCESS CHECK

You can identify the selected KI, active AI provider, project folder button, and project-status button without reading the chat transcript.

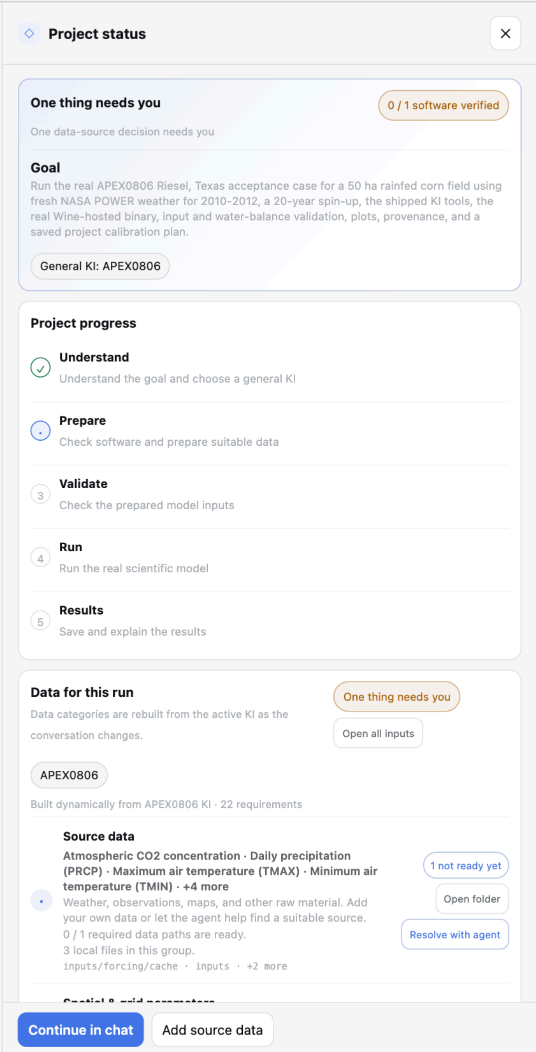
IF IT DOES NOT WORK

If the provider or KI selectors are locked, the chat has already begun. Start a new chat to change the binding; do not silently switch providers halfway through one CLI memory session.

02.07

Open the dynamic project-data panel

Choose Project status. GeoForge reads the selected APEX KI and the current project, then groups only the requirements relevant to this run. It does not show one fixed generic checklist for every model.



WHERE TO LOOK The panel begins with the goal and five-stage progress, then shows current APEX data groups and their real readiness state.

SUCCESS CHECK

The goal mentions this Riesel APEX case, and the data section names APEX-specific requirements rather than a static demonstration list.

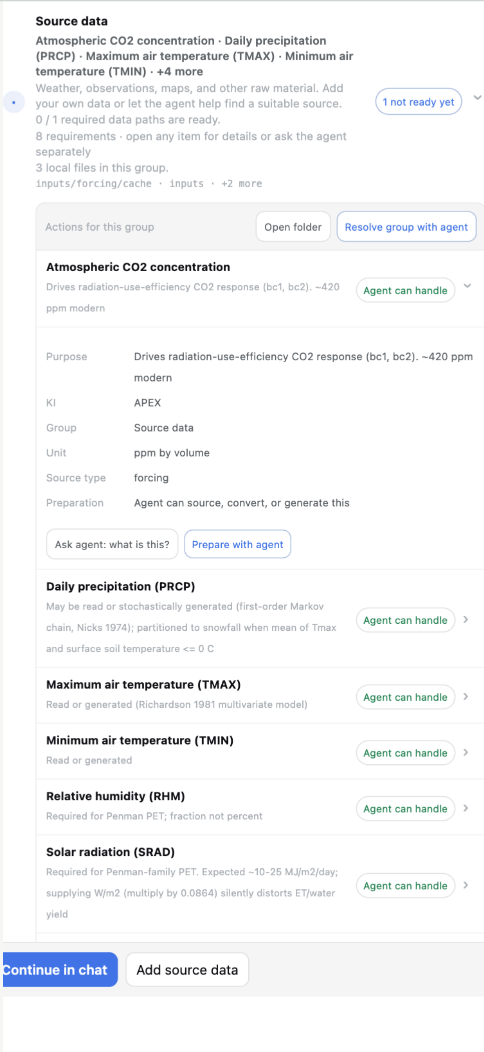
IF IT DOES NOT WORK

If the panel is empty, wait for the first agent response, confirm a KI, then close and reopen Project status. If it still fails, use the chat transcript's error and the project runs folder for diagnosis.

02.08

Inspect and resolve every input requirement

1. Find Source data. In this case it includes daily precipitation, minimum and maximum temperature, atmospheric CO_2, and other APEX weather inputs.
2. Open the category to see every requirement declared by the current KI. Open an individual item to inspect its purpose, KI, unit, expected file, format, source type, default, and valid range.
3. Choose Ask agent: what is this? for a read-only explanation tied to this project. The agent must distinguish item-level evidence from category-level readiness.
4. Choose Prepare/Find/Decide with agent for one item, or Resolve group with agent when the whole category should be handled together.
5. The agent first reuses and validates existing files, then follows the KI to download, convert, derive, or generate. Source URL, request parameters, checksums, units, and transformations stay in the project.



WHERE TO LOOK Categories remain compact, but every KI requirement can be opened, explained by the agent, or resolved separately.

SUCCESS CHECK

Every declared item opens independently, both item buttons work, and Open folder leads to actual files under the current project.

IF IT DOES NOT WORK

Use Add source data for a private file. If a public download is blocked, wait for a structured Needs you request rather than copying an arbitrary replacement into the model directory.

02.09

Respond to a real "Needs you" decision

The example project detects a gap in a public station record. GeoForge does not bury the question in a long AI message. It opens one short decision card: fill the missing period from NASA POWER, or wait for your local station file.

? **Choose the weather data source** Needs you

The public station record has a gap in 2011. Choose how GeoForge should continue; no file will be replaced before you confirm.

- **Use NASA POWER for the missing period**
  The agent downloads the missing dates, converts units, and records the source and checksums.
- **I will add a local station file**
  GeoForge waits for your private file, then validates its fields and dates.

Optional note for the agent

Add context only if needed...

Continue with this choice | Ask about these choices | Not now

WHERE TO LOOK Choose one option, ask the agent to compare them, or postpone. No file is replaced before you confirm.

SUCCESS CHECK

After you choose, the decision is written back into the conversation and the agent continues from the same project state.

IF IT DOES NOT WORK

Choose Ask about these choices. The agent should explain effects on provenance, consistency, and scientific interpretation in simple words, then present the choice again.

02.10

Add files or pin a skill without leaving the chat

1. Choose Files to open the system file picker, or drag files onto the message bar.
2. Choose Skills only when you want to pin a specific reusable procedure to this chat. Otherwise the agent may choose an appropriate skill automatically.
3. Attached files are copied into this project's input area and remain available to later messages in the same project.

+ Files | ✦ Skills | Add a local observation file, or pin a plotting skill to this chat. | Send

WHERE TO LOOK Files, skills, the prompt, and Send remain together at the bottom of the active chat.

SUCCESS CHECK

The file appears in the attachment list before sending, or the Skills button shows the number of pinned skills.

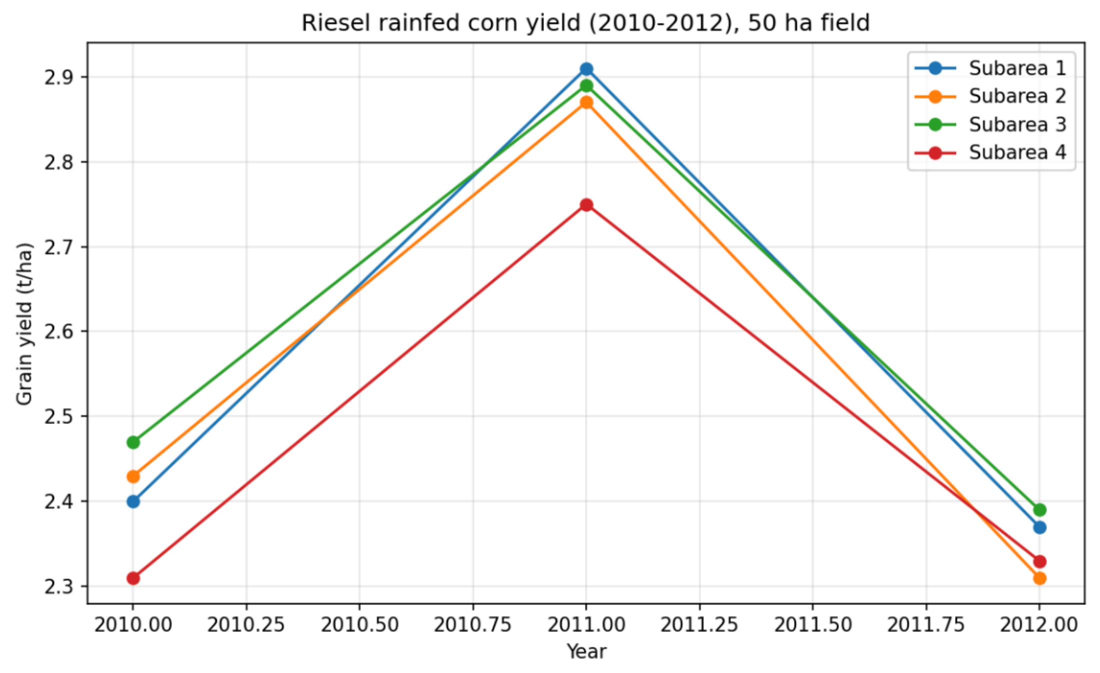
IF IT DOES NOT WORK

If a large file is rejected, copy it into the project inputs folder through Finder and tell the agent its relative path. Keep the original file outside the model's executable directory.

02.11

Verify the run from the saved yield plot

The completed acceptance case produced finite corn yields for 2010–2012. The actual-year range was 2.31–2.91 t/ha, with a mean of 2.54 t/ha. The Riesel reference template retains multiple subareas, so the saved plot contains four series even though the main configured subarea is 50 ha. This is a detail to inspect, not hide.



Riesel annual corn yield

WHERE TO LOOK This image is rendered from a file under artifacts, and the underlying annual CSV remains in the project.

SUCCESS CHECK

The plot is visible in chat, opens as a saved artifact, and its values agree with the parsed CSV and the assistant's summary.

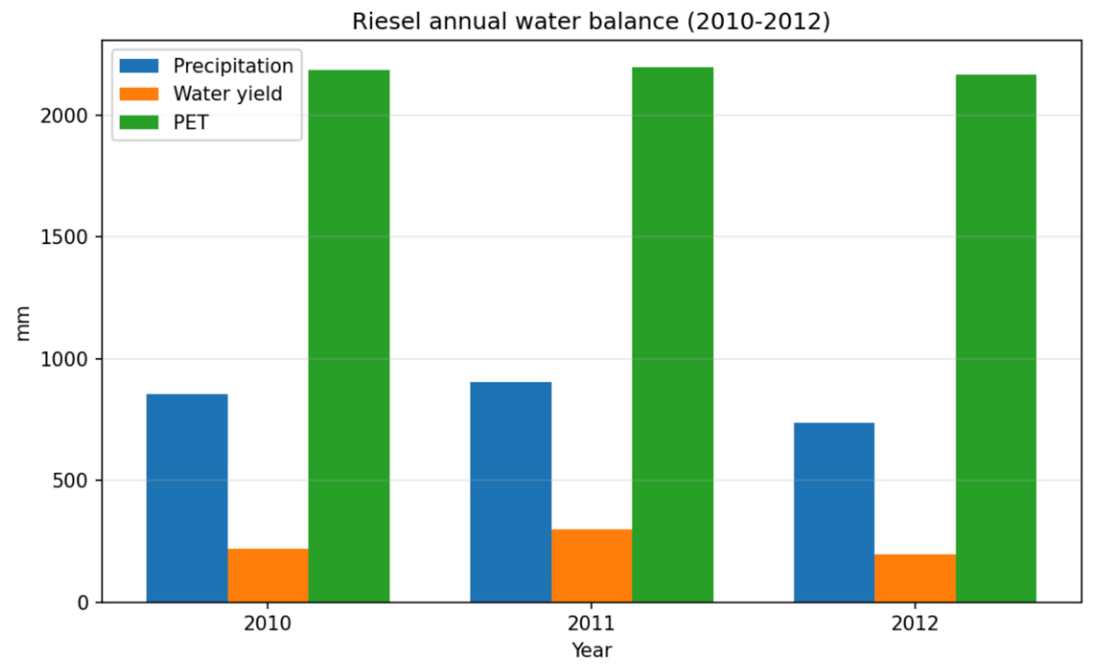
IF IT DOES NOT WORK

If a plot appears without a saved data table or model output, do not accept it as evidence. Ask the agent to identify the source file, parser, and exact artifact path.

02.12

Check the water balance, not only the crop plot

The case closed its water balance to 0.47% using precipitation against runoff, evapotranspiration, deep percolation, subsurface flow, and storage change. A small closure error supports internal consistency; it does not by itself prove that every input or parameter is scientifically correct.



Riesel annual water balance

WHERE TO LOOK Use the plot together with the annual water-balance CSV and the reported closure calculation.

SUCCESS CHECK

The result reports the closure equation and value, all plotted values are finite, and the source CSV is saved.

IF IT DOES NOT WORK

Check units, date alignment, spin-up exclusion, missing-value handling, and whether storage change was included before changing model parameters.

02.13

Open the project folder and audit the evidence

1. Choose Folder in the chat header. Finder opens this conversation's project directory.
2. Open inputs to inspect source and prepared data; open outputs/APEX for the runnable model workspace and raw model outputs.
3. Open artifacts for parsed CSV files and plots; open runs for orchestration, progress, and execution records.
4. Open the provenance JSON and confirm the NASA POWER URL/query, checksums, transformations, executable, dates, and model version.

EXPECTED PROJECT STRUCTURE

```
project/
├── inputs/                 source and prepared inputs
├── outputs/APEX/           runnable APEX workspace and raw outputs
├── artifacts/              annual CSV files, yield plot, water-balance plot
├── runs/                   orchestration and execution records
├── calibration/cases/      saved calibration plan
├── memory/                 durable project memory
└── session.json            conversation and binding metadata
```

SUCCESS CHECK

Another researcher can identify the exact input, executable, command/workflow, raw output, parsed table, plot, and provenance record without relying on the chat prose.

IF IT DOES NOT WORK

If the chat says a file exists but Finder cannot find it, ask for the absolute project path and relative artifact path. Treat the file system as the source of truth.

02.14

Understand what the saved calibration plan means

This run saved a calibration case plan under calibration/cases with targets such as yield and water balance, candidate parameters, an optimization strategy, and a holdout design. The APEX adapter was still marked authoring-needed, so the calibration engine did not run. GeoForge correctly reported a plan, not a calibrated or transferable model.

- ✓ Plan saved: the workflow can be reviewed and completed later.
- ✓ Calibration run: requires a working project-specific adapter and real observations.
- ✓ Calibrated: requires successful model evaluations and objective results.
- ✓ Validated/promotable: additionally requires independent holdout validation to pass.

SUCCESS CHECK

The project contains a readable case plan, but the result does not claim calibration unless engine records and holdout metrics exist.

IF IT DOES NOT WORK

Add real observations, then use the calibration section in Project status to have the agent complete a project-local adapter. Never edit the shared engine or original KI to make one project pass.

NOTE The honest stopping point is part of reproducibility. A saved plan is useful, but it is not a calibration result.

03

APP INSTALLATION, GATEKEEPER, AND LOCAL STORAGE

Install and launch on macOS

GeoForge is a local desktop application. The app bundle contains the interface and shared harness; individual scientific models may still require installation, licensing, or a separate download.

03.01

Install the app

1. Download the macOS release that matches your Mac. Apple Silicon is arm64; older Intel Macs use x86_64 where offered.
2. Unzip the release and move GeoForge Desktop.app to Applications.
3. On first launch, macOS may ask for confirmation. Use System Settings › Privacy & Security only if macOS identifies the downloaded app and you trust the release source.

03.02

Choose where work is stored

Every new conversation proposes a folder inside the GeoForge projects directory. You can accept it or choose another local folder. Keep projects outside the application bundle so updates never replace your work.

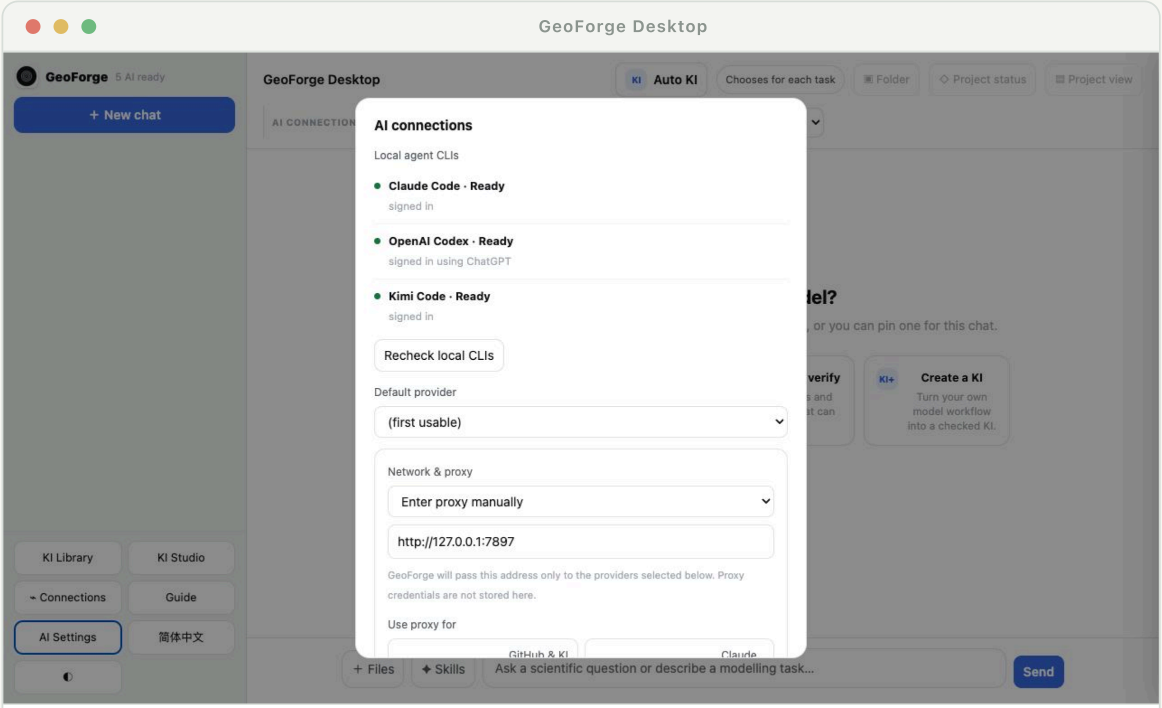
> **NOTE** For large models, choose a disk with enough free space. Avoid cloud-synced folders while a model is actively writing many files.

04

CLAUDE CODE, CODEX, KIMI, AND DEEPSEEK

Connect an AI provider

GeoForge can use an authenticated local CLI or an API. The same KI harness and project tools sit underneath both; the provider supplies the reasoning and language interface.



PRODUCT SCREENSHOT

AI Settings combines CLI readiness, default provider, network routing, Kimi access, API keys, and a complete route test.

04.01

Local CLI mode

- ✓ Claude Code: install and sign in in Terminal, then use Recheck in GeoForge.
- ✓ OpenAI Codex: install/update the CLI and complete its normal sign-in flow.
- ✓ Kimi: install Kimi Code, sign in, and select it from the local provider list.
- ✓ GeoForge starts the provider with the same user account but a controlled project working directory.

04.02

API mode

For DeepSeek or another supported API, enter the key in AI settings or expose it through the supported environment variable. A real one-message probe confirms that the key, endpoint, model, and network path work together.

IMPORTANT Installed does not mean signed in. Signed in does not mean the selected model is available. GeoForge reports these states separately.

04.03

Route only the services that need a proxy

1. Open AI Settings and choose Use Mac system proxy, Enter proxy manually, or Do not use proxy.
2. For a manual proxy, enter the local address, for example http://127.0.0.1:7897. GeoForge does not store proxy credentials; use a local authenticated proxy when credentials are required.
3. Enable the route separately for GitHub & KI updates, Claude, Codex, Kimi, or an API provider. Kimi and DeepSeek do not need to inherit a proxy merely because Claude does.
4. The selected provider's agent-run Git, pip, curl, and download commands use the same route. Choose Test AI & GitHub before starting a long setup.

SUCCESS CHECK

The provider probe and GitHub probe both report a real successful connection through the intended route.

IF IT DOES NOT WORK

If direct access worked before but now fails, compare the selected mode with the Mac system proxy and test the proxy host itself. A DNS error, timeout, sign-in error, and unavailable model are different failures.

04.04

Choose Kimi Code file access

- ✓ Project scoped is recommended. Kimi can use the current project, selected KI, approved data paths, and its own login/session state; other personal files remain blocked.
- ✓ When Kimi requests a required external folder such as its user skills directory, GeoForge opens a focused permission dialog. Choose Allow once or Always for this project rather than granting the whole home folder.
- ✓ Full computer access is a compatibility fallback. Use it only when project-scoped execution cannot work and after reviewing the risk.

NOTE A permission dialog should appear once for a new unresolved path. If it repeats after Always for this project, inspect the saved project policy instead of approving it over and over.

04.05

What is sent to a provider

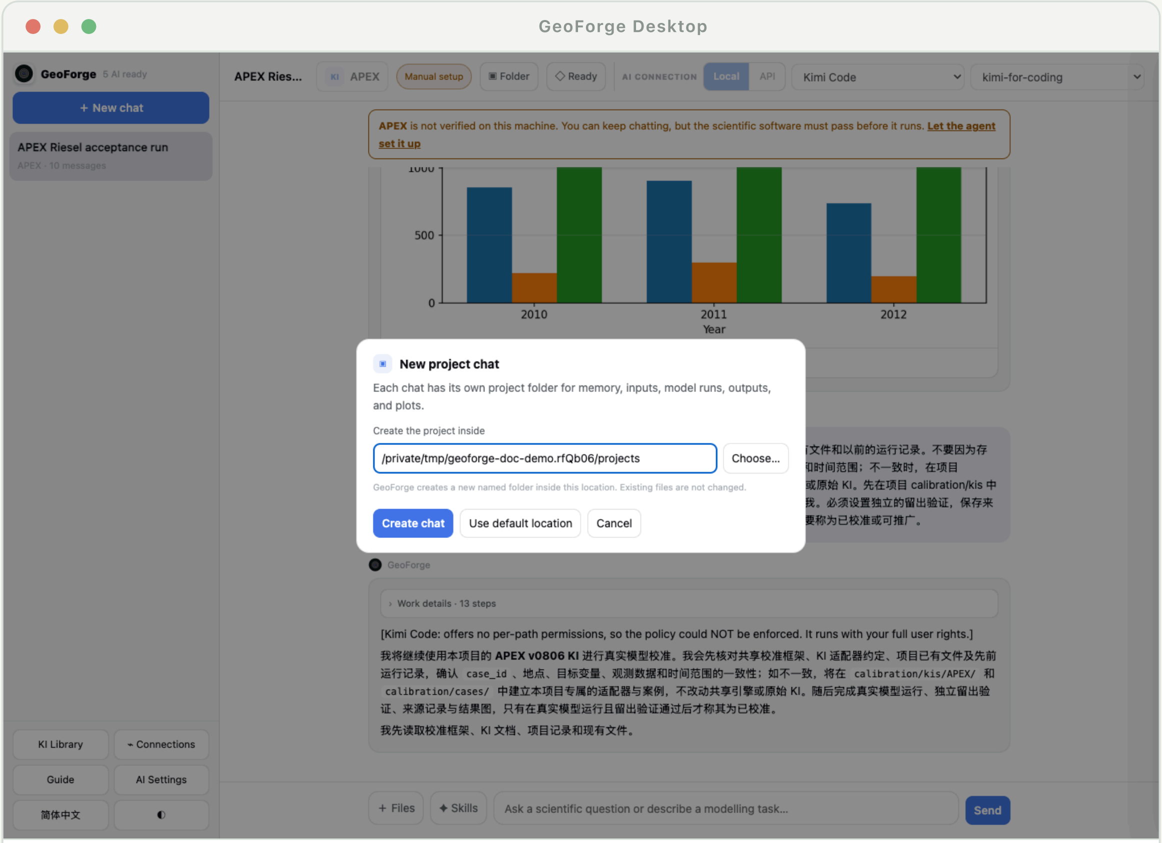
The conversation, relevant KI instructions, and selected tool results may be sent to the active AI provider. Local model files are not automatically uploaded in full; the agent reads only what a tool exposes for the task. Review provider terms before using confidential data.

05

ONE DURABLE LOCAL FOLDER PER CONVERSATION

Projects, conversations, and memory

GeoForge does not depend on a provider remembering your project forever. Each conversation has a local project folder and a durable transcript. The provider's native session ID is kept when supported, but local project state remains the source of truth.



PRODUCT SCREENSHOT

The default path is visible and can be changed before the chat begins.

05.01

What the project folder contains

- ✓ Conversation history and concise project state used to resume work.
- ✓ Selected KIs, execution policy, provider/session metadata, and progress events.
- ✓ Source files, prepared inputs, model workspaces, outputs, plots, and reports.
- ✓ Provenance: where data came from, how it was transformed, and which command produced an artifact.

05.02

Switching conversations while work continues

A running conversation should show visible activity and continue in its own task. You can create or open another conversation without treating the whole app as frozen. Returning to the original project shows its latest event, pending request, or result.

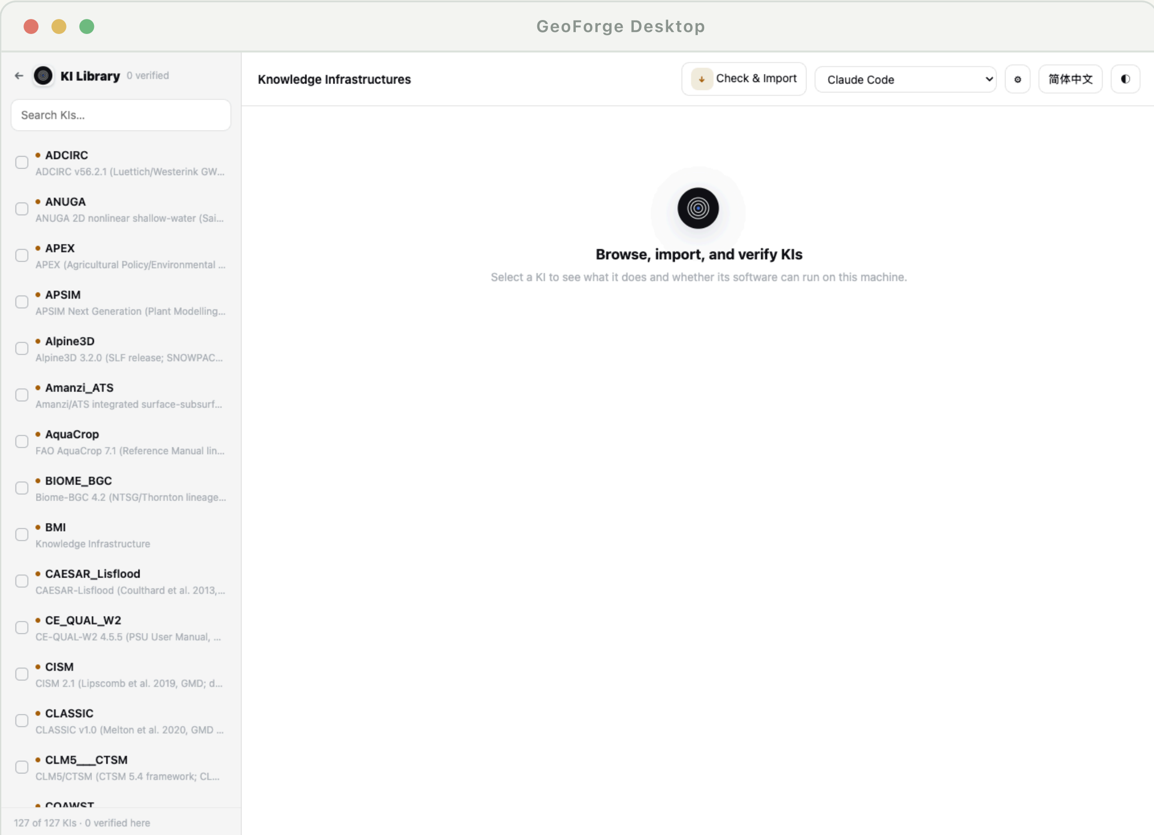
> **TIP**
>
> Use one conversation for one scientific objective. Start a new project when the question, study area, or model experiment changes materially.

06

AVAILABILITY, VERIFICATION, IMPORT, AND TRUST

Understand the KI Library

A Knowledge Infrastructure (KI) is the model-specific operating knowledge used by the agent. It can include setup recipes, tools, diagnostics, data contracts, examples, documentation summaries, and interpretation rules.



PRODUCT SCREENSHOT

A KI can be present before its underlying model has been verified on this Mac.

06.01

Three states that must not be confused

- ✓ KI available: GeoForge can read the model guidance and tools.
- ✓ Software verified on this Mac: the required executable passed the KI's real preflight check here.
- ✓ Project ready: this particular scenario has enough validated inputs and decisions to run.

06.02

Import and verify a KI

1. Choose Import KI and select a KI package or repository folder.
2. Run the verifier. It checks structure, declared tools, paths, and safety rules before import.
3. Review the scientific software, version, license, source, and last-check result.
4. Use Setup & verify when the KI exists but its model software is not yet ready.

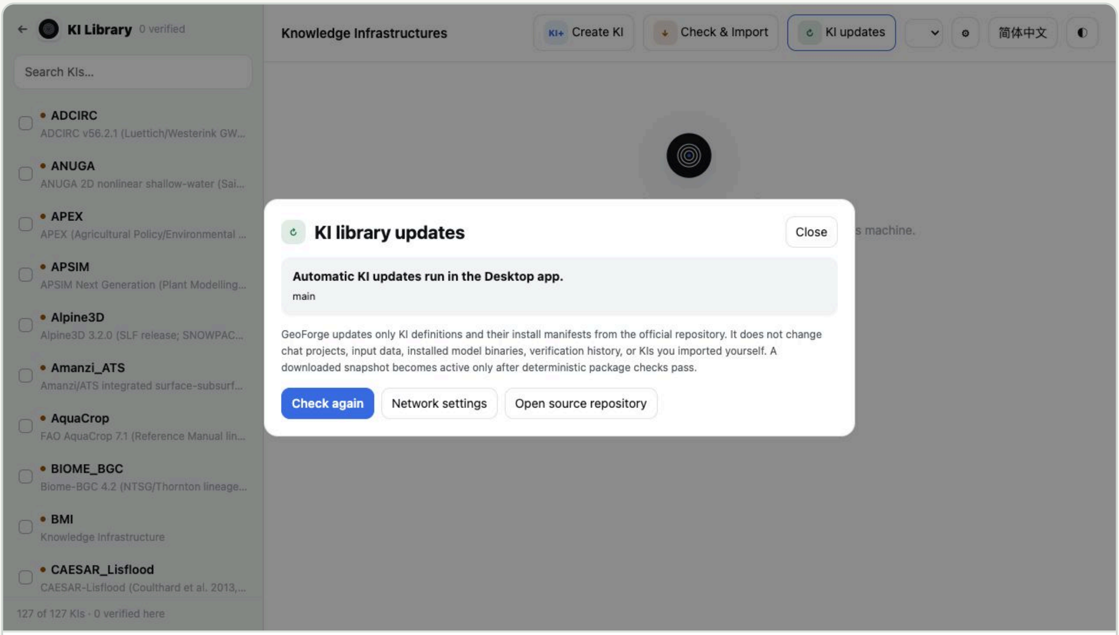
> **NOTE** Verification is local and version-specific. A model verified on a Linux server is not automatically verified on macOS.

06.03

Automatic KI updates from the official main branch

At launch, GeoForge checks the official repository's main branch for model KI definitions and install manifests. A downloaded snapshot becomes active only after deterministic package checks pass. If the network or validation fails, the last validated snapshot remains active.

- ✓ The updater may add, change, or remove official KI files and reports the branch, time, route, and summary.
- ✓ It never overwrites chat projects, input data, installed model binaries, verification history, or KIs you imported yourself.
- ✓ Use the GitHub & KI updates proxy switch when the repository cannot be reached directly.



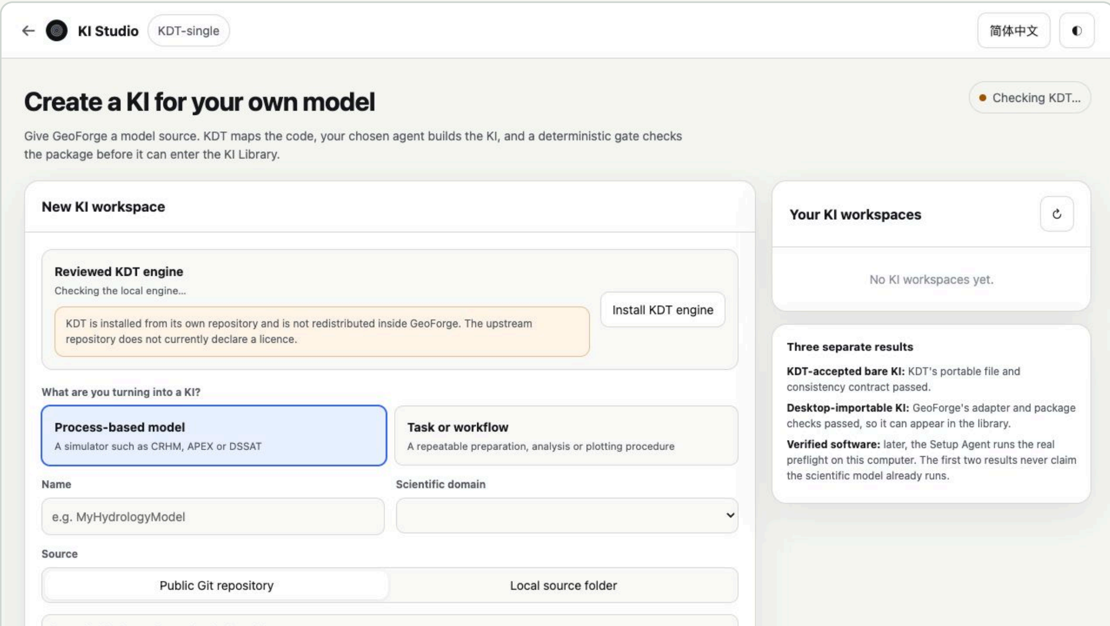
WHERE TO LOOK The report states the source branch and exactly what the updater is allowed to change.

06.04

Create a KI in KI Studio

KI Studio runs the reviewed KDT-single engine against a public Git repository or a local source folder. It can map a process-based model or a repeatable task/workflow. Choose a guided scientific domain or enter your own; an unguided domain starts without prior KDT domain knowledge and must be verified from the source.

1. Create an isolated workspace and run KDT source discovery.
2. Choose or change the AI provider for construction and later repair passes.
3. Provide only genuinely missing protected papers, documentation, credentials, or licensed assets when the request panel asks.
4. Run the KDT gate, then GeoForge's Desktop package/import gate. Model setup and real preflight occur later on the target computer.



WHERE TO LOOK The three result levels prevent a structurally valid bare KI from being mistaken for installed scientific software.

IMPORTANT KDT-accepted bare KI, Desktop-importable KI, and Verified software are separate results. The first two never prove the scientific executable runs.

07

WHAT TO ASK AND WHAT THE AGENT DECIDES

Work through chat

Users should not have to translate a scientific goal into dozens of file names. Start with the question. The agent uses the KI to discover technical requirements and surfaces only decisions that cannot be safely inferred.

07.01

A good first message

Include the process, place, period, scale, and desired comparison or output when you know them. Example: “Simulate rainfed maize yield around Xinxiang for 2000–2020 and compare a warmer scenario with the baseline.” It is fine to omit items you have not decided.

07.02

What the agent should handle

- ✓ Select or propose a suitable KI and explain the choice briefly.
- ✓ Inspect software readiness and repair ordinary setup problems inside the model workspace.
- ✓ Turn the KI data contract into a short project-specific data plan.
- ✓ Download public data, transform formats, generate derivable parameters, and validate inputs where permitted.
- ✓ Run the model, check outputs, make plots, and explain uncertainty without hiding failures.

07.03

When GeoForge should ask you

- ✓ A scientific choice has several defensible answers that change interpretation.
- ✓ A private file, login, license, protected download, or macOS permission is required.
- ✓ An action would write outside the project, install system software, or access a sensitive location.

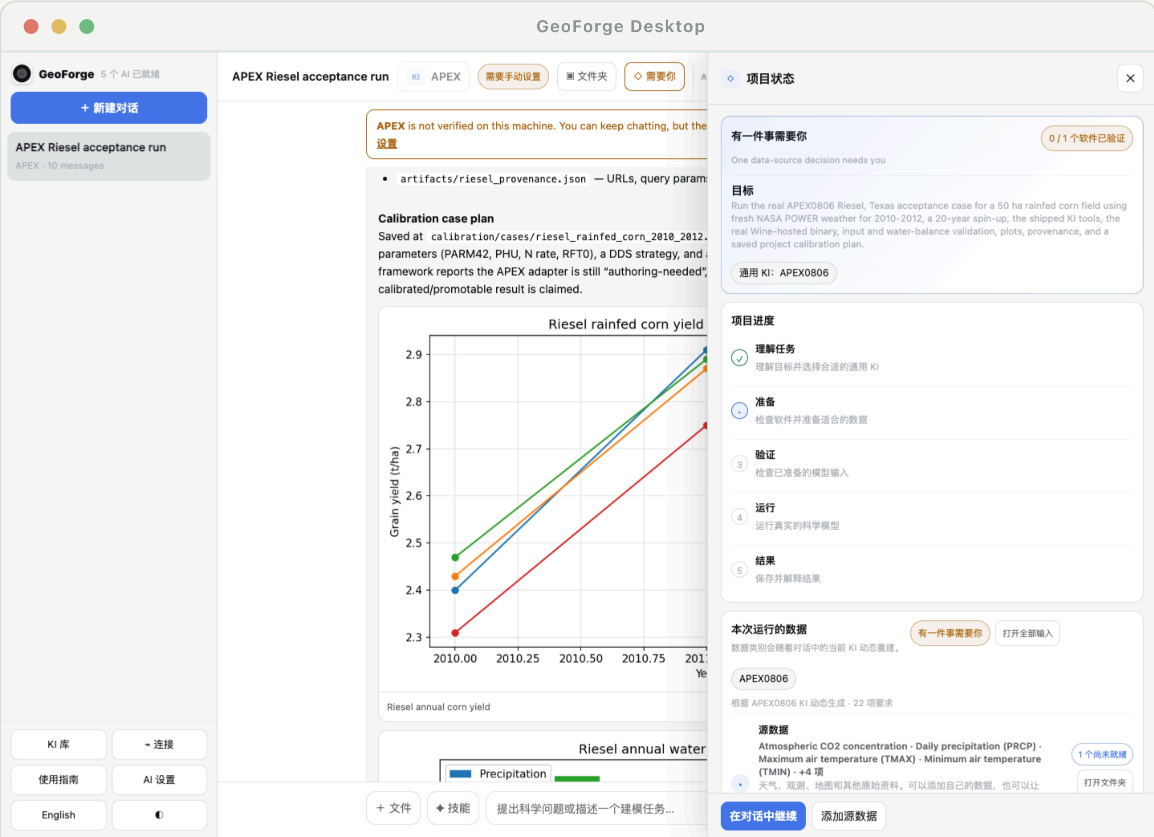
> **TIP** You can always ask: “What did you assume, what is measured, and what is generated?”

08

A DYNAMIC, KI-SPECIFIC VIEW OF WHAT THIS RUN NEEDS

Prepare project data

The data panel is generated after the agent understands the scenario and selects one or more KIs. It is not a static checklist and should not expose every internal model file as work for the user.



PRODUCT SCREENSHOT

The overview stays compact; technical requirements remain available under details.

08.01

The four compact lanes

- ✓ Source data: weather, observations, maps, measurements, or other raw material.
- ✓ Spatial and grid parameters: catchments, layers, cells, HRUs, reaches, or point locations derived from source data.
- ✓ Model choices and calibration: parameters, objective functions, bounds, and defensible defaults.
- ✓ Initial and boundary conditions: the state at the start and the conditions around the modeled domain.

08.02

Use Resolve with agent

1. Open a category to see its current state: ready, working, needs review, or needs you.
2. Choose Resolve with agent. The agent reads the selected KI contract and the current scenario.
3. It finds permitted public sources, converts formats, derives technical files, and records provenance.
4. If a real choice or permission is needed, the task moves to Needs you with a focused request.

08.03

See the actual files

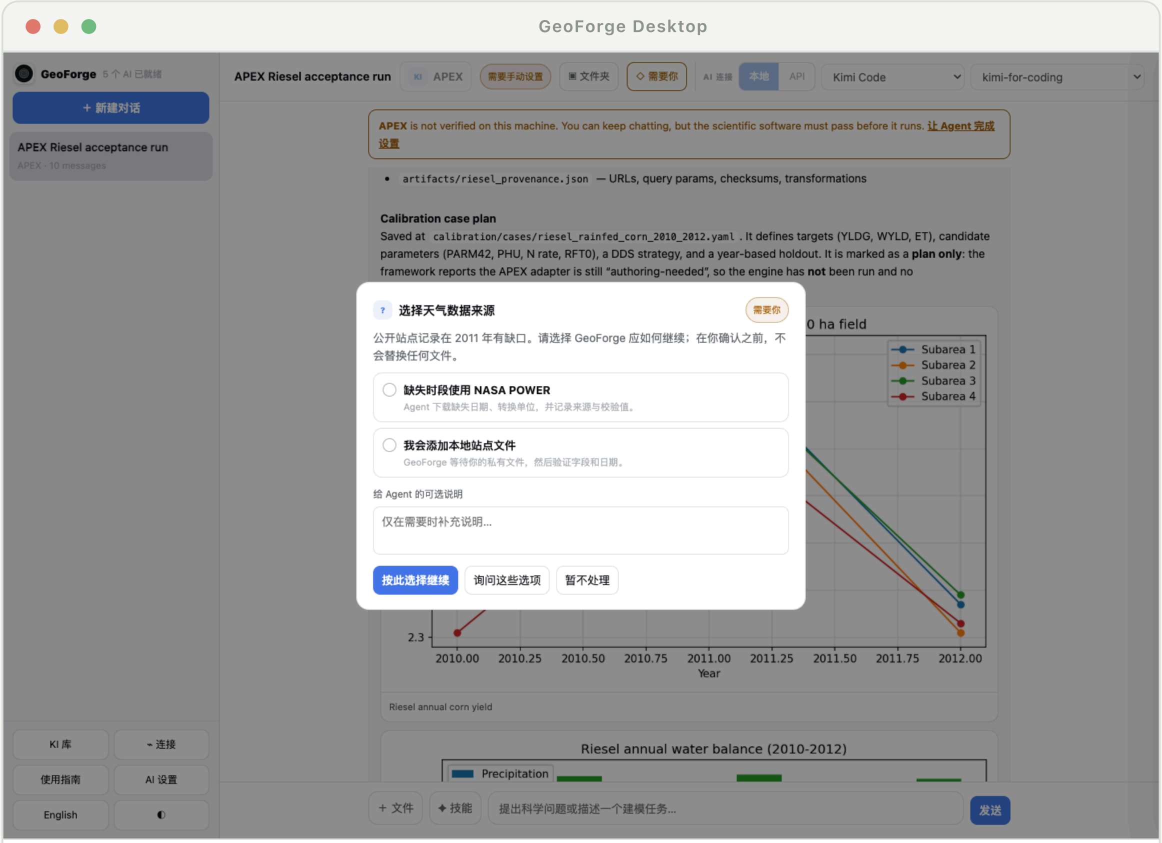
Use Open folder on the panel or the Folder button in chat. GeoForge opens the project location in Finder so you can inspect the source, prepared input, and output files directly. Detailed entries should show local path, source URL or citation, transformation, validation, date, and checksum when available.

> **IMPORTANT** Ready means the requirement has a validated project path. It does not mean every value is observed or uncertainty-free.

DECISIONS, PERMISSIONS, LICENSES, AND MANUAL DOWNLOADS

Respond to “Needs you”

When the agent cannot safely continue alone, GeoForge should stop with a structured request—not bury a command inside a long chat message. The request explains what is blocked, why it matters, and the smallest action that unlocks progress.



PRODUCT SCREENSHOT

Choose a safe action, decline it, or return to chat to discuss another route.

09.01

Common request types

- Permission: allow one file, folder, executable, or clearly scoped command.
- Download: open a publisher or data portal that requires terms, CAPTCHA, or an account.
- License: confirm that you have the right to install or run restricted software.
- Scientific decision: choose among alternatives whose consequences the agent explains.
- Private input: select a local file without making it public or silently substituting synthetic data.

09.02

Dynamic permission policy

GeoForge can grant narrowly scoped, recoverable actions during a project. A useful pattern is Allow once, Always allow for this project, Choose another file, and Deny. Broad access to the home directory or unrestricted shell execution should never be the default.

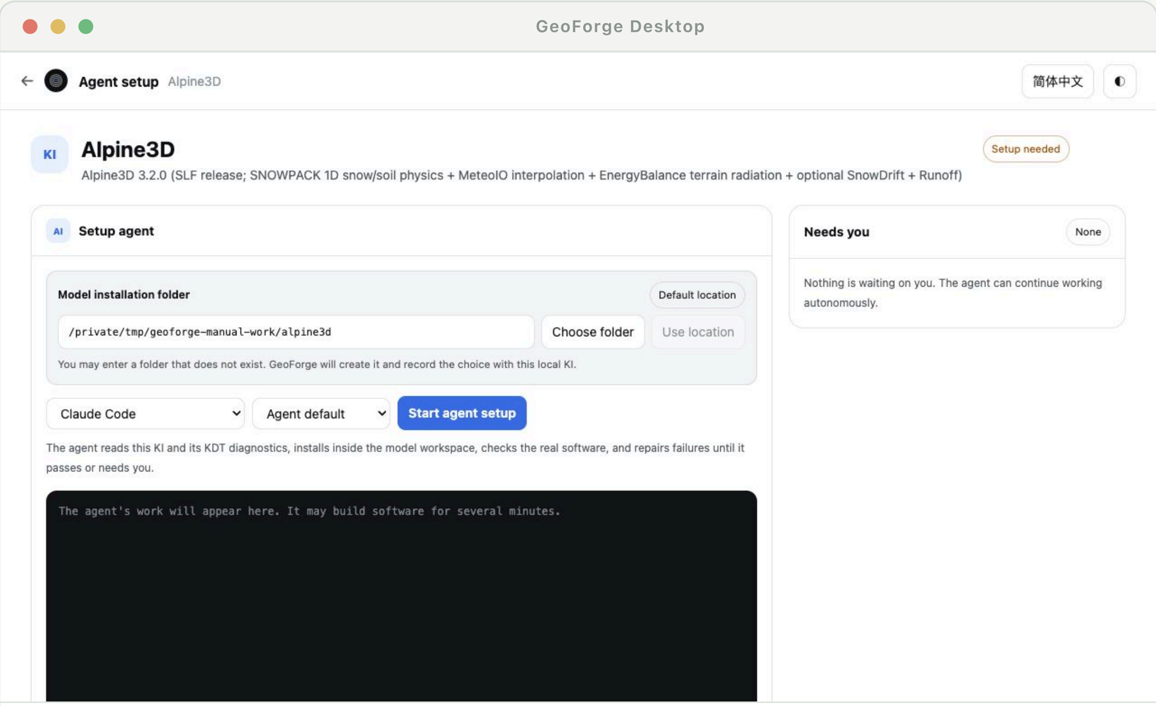
> **TIP** If an option is unclear, choose Discuss in chat. The agent should explain the risk and find a project-local alternative when possible.

10

AGENT-LED INSTALLATION WITH HONEST CHECKPOINTS

Set up and verify model software

The setup agent follows the KI's KDT recipe in a model workspace. It materializes tools, prepares an environment, acquires or builds the correct version, and runs a real preflight. Ordinary failures should be diagnosed and retried automatically.



PRODUCT SCREENSHOT

Choose the installation folder before the setup agent starts; GeoForge can create a path that does not exist yet.

10.01

Choose and record the installation folder

1. Accept Default location, choose an existing folder, or type a new folder path. The folder does not need to exist yet; GeoForge creates it after validation.
2. Use one stable model workspace rather than putting software inside a temporary chat folder.
3. GeoForge records the chosen location in the installation configuration and the local KI install record, so later chats and preflight checks resolve the same path.
4. Portable KI placeholders are resolved at runtime. A KI must not depend on a developer's hard-coded /Users/... or Windows drive path.

> **TIP** Changing the field does not move an existing installation automatically. Choose the recorded installed location or run an explicit migration.

10.02

What verification proves

1. The KI was materialized and its required shared tools are importable.
2. Required runtime and system dependencies are present.
3. The intended software version was obtained or built for this platform.
4. The KI's real preflight starts the executable and checks expected behavior.

10.03

What the agent repairs itself

- ✓ A Git branch/tag mismatch, retryable clone failure, build directory, or project-local path.
- ✓ A missing project-local Python package such as ki_tools_common.
- ✓ A model binary that must be copied or linked into the controlled workspace.
- ✓ A preflight failure with a documented KI repair path.
- ✓ The setup loop has no artificial step count. It may continue through legitimate repair passes until verification succeeds, a real user action is required, or the provider stops.

10.04

Read the live activity state

- ✓ Connected: the provider process is alive and GeoForge has received a recent event.
- ✓ Working: the visible stage and step count are changing; long builds may legitimately take several minutes.
- ✓ Waiting: the provider is alive but waiting for a response, permission, download, or model command.
- ✓ Possibly stalled: no heartbeat or new event arrives within the stated threshold. GeoForge should offer diagnosis or cancellation without pretending that it is still progressing.

> **NOTE** Elapsed time alone does not prove a hang. The heartbeat, last event, process exit state, and provider error together determine the status.

10.05

What still needs you

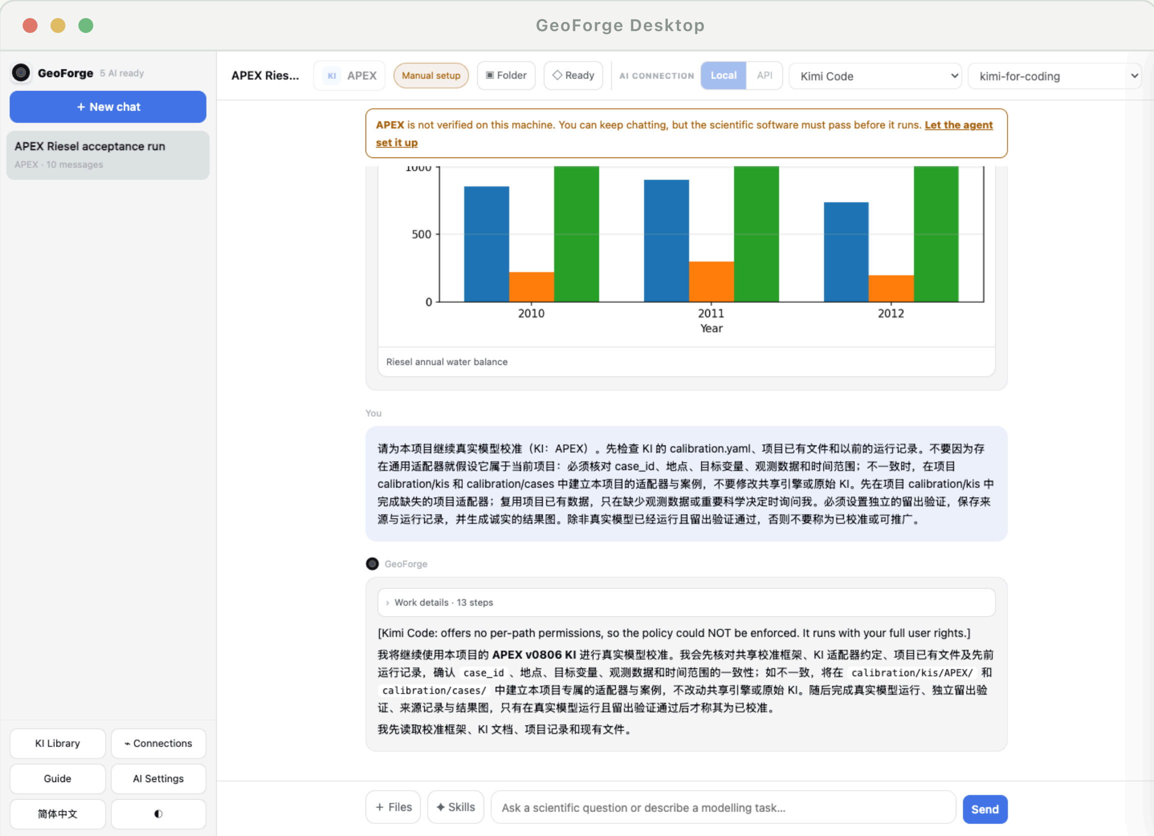
The agent must stop for licenses, protected downloads, credentials, macOS system permissions, or destructive/system-wide changes. Software verification remains separate from project data: missing weather or DEM data should not label a successfully installed model as broken.

> **IMPORTANT** A setup log ending with “Load failed” is not enough. The UI should preserve the actual provider error, exit status, and last successful step.

FROM PREPARED INPUTS TO DEFENSIBLE RESULTS

Run, validate, and interpret

A successful GeoForge run has three layers: execution evidence, validation evidence, and scientific interpretation. The agent should make each layer visible and save it with the project.



PRODUCT SCREENSHOT

Results remain linked to project artifacts and validation checks.

11.01

Before execution

- ✓ Confirm study extent, dates, model version, units, and scenario assumptions.
- ✓ Validate required paths, file formats, missing values, ranges, coordinate reference systems, and temporal coverage.
- ✓ Freeze or record the prepared inputs so the run can be reproduced.

11.02

After execution

- Check the process exit status and model-specific completion markers.
- Reject empty, stale, truncated, impossible, or non-finite outputs.
- Run mass balance, benchmark, range, continuity, or reference-case checks declared by the KI.
- Separate simulated results from observations, assumptions, and calibration targets.

11.03

Read the answer critically

A model result is conditional on the selected structure, data, parameterization, and scenario. Ask for sensitivity, holdout performance, missing processes, and the main reasons the result could change. GeoForge should report a failure or limitation directly rather than constructing a scientific story around an incomplete run.

SEE EVIDENCE IN CHAT AND KEEP THE REAL ARTIFACT

Plots, tables, and project files

The agent can render plots and compact tables directly in chat. Every important visual should also be saved in the project so it can be reopened, regenerated, or used in a report.

12.01

Inline visualization

- ✓ Plots should state variables, units, dates, spatial scope, and whether values are observed or simulated.
- ✓ Calibration plots should distinguish calibration and validation periods.
- ✓ Maps should state projection, resolution, extent, and masking rules.
- ✓ A visual error should not erase the underlying model outputs; the agent can repair and re-render it.

12.02

Open the dynamic Project View

Project View is rendered per conversation from a safe project manifest. The agent and selected skills can publish plots, tables, maps, time controls, and model-specific views such as a 2-D flood sequence or BIM scene. A later specialized KI can define its own view while keeping the underlying artifacts in the project.

- ✓ The view is not one static dashboard shared by every model.
- ✓ Only declared, project-contained artifacts and supported view components are rendered; arbitrary agent HTML or JavaScript is not executed.
- ✓ Use the project folder to inspect the source data behind every visual and regenerate it later.

12.03

Add your own files

Drag files onto the message composer or choose + Files to open the file picker. GeoForge copies or references them inside the project according to the selected policy, records their original names, and tells the agent where they are. Large folders should be added through the folder picker instead of pasted into chat.

> **TIP** Use Copy on a message for text, and Open folder for artifacts. Copying a displayed path is not the same as opening or granting access to it.

13

SHARED ENGINE, KI ADAPTER, AND PROJECT CASE

Build a reusable calibration workflow

Calibration is not a one-off conversation. GeoForge turns it into a saved workflow: a shared calibration engine, a model-specific KI adapter, and a project-specific case that can be rerun after changing bounds, observations, or budgets.

13.01

The three layers

- ✓ Calibration engine: algorithms, scheduling, checkpoints, metrics, and result storage shared across models.
- ✓ KI adapter: how parameters are written, how the model is launched, and how outputs become objective values.
- ✓ Project case: this study's parameters, bounds, observations, warm-up, calibration/validation periods, objective, and compute budget.

13.02

Create the workflow

1. Ask to calibrate after a baseline run succeeds and the observed target is validated.
2. Review proposed parameters, ranges, constraints, objective metrics, and holdout design.
3. Generate the adapter and case in the project's calibration folder.
4. Run a smoke test, then the selected budget. Failed candidates remain traceable rather than disappearing.
5. Save the best parameter set, full history, diagnostics, and a command or button for rerunning.

13.03

Update later

Open the saved calibration case, adjust allowed values such as iteration budget, parameter bounds, or target files, and choose Run calibration. GeoForge should create a new run record instead of overwriting the earlier result. Model or adapter version changes require a new validation check.

> **IMPORTANT** Calibration can improve fit without improving truth. Always reserve independent data or a validation period and report both performances.

ADD SPECIALIST METHODS AND CONTROLLED EXTERNAL TOOLS

Skills and MCP connections

KIs explain how to operate scientific models. Skills add reusable methods such as plotting, statistical analysis, geospatial processing, or literature workflows. MCP connections give controlled access to external services such as GitHub.

14.01

Use skills in a conversation

Choose Skills beside the message box to review or pin skills for that conversation. You can also type a slash command or ask naturally, such as “make a publication-quality hydrograph.” The agent may automatically use a relevant installed skill, while the selected list makes your intended methods explicit.

14.02

Connect MCP services

1. Open Connections and choose an available MCP provider.
2. Review what resources and actions it exposes before authenticating.
3. Complete login in the provider's trusted flow and return to GeoForge.
4. Grant only the repositories, folders, or scopes needed for the project.

> **NOTE** A connection expands what the agent can access; it does not replace KI validation or project provenance.

A CONSISTENT ENGLISH OR SIMPLIFIED CHINESE EXPERIENCE

Language, copy, and keyboard use

The interface language controls labels and guides. The conversation follows the language of your latest message, so a Chinese request should receive a Chinese answer unless you ask otherwise.

15.01

Language behavior

- Use English/中文 in the app to switch navigation, status, dialogs, and this guide.
- Scientific names, model file names, commands, units, and paths remain unchanged when translating would make them incorrect.
- Ask “answer in English” or “请全程使用中文” to override automatic conversation language.

15.02

Copy, paste, and shortcuts

- Select text normally and use Command-C / Command-V in the composer.
- Use Copy on an assistant message to copy its rendered text without internal tool markup.
- Press Command-K for a new task where shown; Enter sends and Shift-Enter inserts a new line.
- While an agent is running, navigation and New conversation remain available.

KNOW WHERE DATA GOES AND PRESERVE THE SCIENTIFIC RECORD

Privacy and reproducibility

GeoForge is local-first, but the selected AI provider and connected services may receive task context. Reproducibility also requires more than keeping outputs: the project must preserve the exact inputs and decisions that produced them.

16.01

Before using sensitive data

- ✓ Check the active provider, its data policy, and whether a local CLI still sends prompts to a cloud service.
- ✓ Remove secrets, personal identifiers, unpublished restricted data, and credentials from chat and ordinary project files.
- ✓ Use project-scoped permissions and disconnect external services when they are no longer required.
- ✓ Do not upload licensed papers or model binaries unless their terms permit it.

16.02

Minimum reproducibility record

- ✓ GeoForge, KI, model, adapter, and calibration-engine versions.
- ✓ Source data identifiers, URLs/citations, access dates, checksums, and licenses.
- ✓ All transformations, units, projections, missing-data decisions, and generated parameters.
- ✓ Execution commands, environment, random seeds, run dates, failures, and validation checks.
- ✓ Plots and tables linked to the exact result files they summarize.

> **TIP** Archive or export the whole project when a result supports a paper, report, or decision.

17

PLAIN FIXES FOR THE PROBLEMS USERS SEE MOST

Troubleshooting and glossary

Start with the visible status and project event log. Do not repeatedly restart a task without reading the last real error: it may be waiting for authentication, permission, a download, or a scientific decision.

17.01

Common problems

- CLI says sign-in needed: run the provider's own status command in Terminal, confirm the same macOS user, then Recheck. A stale probe must not be treated as truth.
- The app looks frozen: check the activity badge and project events. API calls and model runs should stream a stage, elapsed time, and Cancel/continue-in-background option.
- No module named ki_tools_common: rematerialize shared tools into the project environment and verify the tool path before retrying.
- Preflight failed because data is missing: distinguish software preflight from project-input validation. Verify software first, then prepare scenario data.
- Git clone failed: the setup agent should inspect available tags/branches, network status, and partial directories, then retry safely. Ask you only for authentication or protected access.
- Copy/paste does not work: click inside the composer, use the Edit menu or Command-C/Command-V, and report the exact field if selection remains blocked.

17.02

Glossary

- KI — Knowledge Infrastructure: model-specific knowledge, tools, checks, and data contracts used by the agent.
- KDT — the executable setup/diagnostic recipe that makes a KI's software usable on a machine.
- Preflight — a real check performed before a full project run; it should prove the intended software can execute.
- Project run — one traceable execution with fixed inputs, configuration, outputs, and validation.
- Adapter — the model-specific bridge between a shared framework and one model's files, command, and outputs.
- Holdout — observations or a period excluded from calibration and used for independent evaluation.
- Provenance — the recorded origin and transformation history of data, parameters, code, and results.

17.03

Report a useful bug

Include the GeoForge version, macOS version and chip, active provider/model, KI, project stage, exact visible error, and the relevant project event/log. Remove API keys and private data. A screenshot is useful, but the underlying text and timestamp are usually more diagnostic.

GeoForge

Turn scientific knowledge into runnable, inspectable, reproducible workflows.

Updated 27 August 2026

GEFORGE DESKTOP · PRODUCT BOOK

GeoForge 桌面版使用手册

让科学模型
真正跑起来。

从科学问题到可复现的模型运行. 本手册从用户真正看到的界面出发，说明每一步在做什么、哪些由 Agent 完成、哪些决定仍属于你。

一次可复现的运行 01—05

目录

00 只需记住三件事

✦

对话是主入口

用自然语言描述科学任务。GeoForge 会把它变成可见、可检查的工作流。

KI

KI 是可执行的科学指导

KI 告诉 Agent 如何安装、准备、运行、检查并解释一个科学模型。

⌂

每个项目都保存在本地

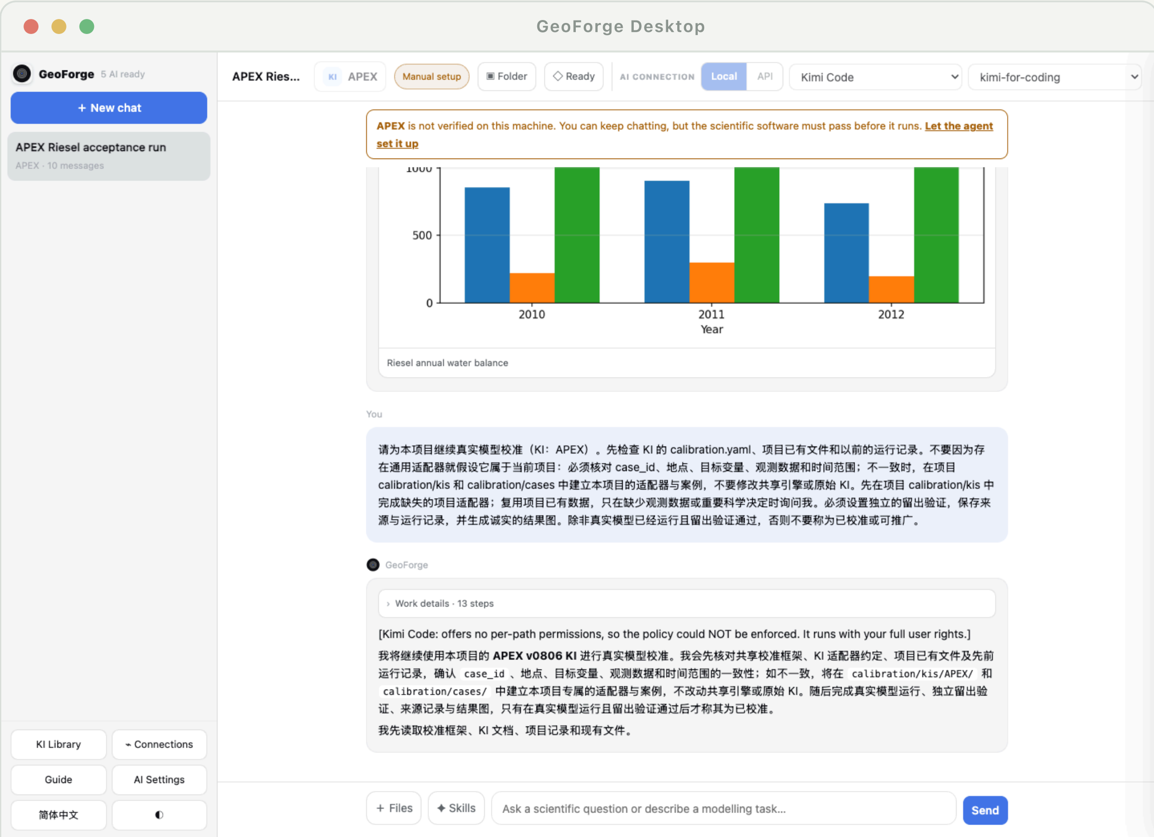
消息、源数据、模型输入、输出、图表和来源记录都保存在 Mac 上同一个项目文件夹中。

01

10 分钟完成一次 GEOFORGE 工作流

第一次成功运行

最短路径是：新建一个本地项目，描述具体情景，让 Agent 自动完成可以安全完成的准备，只回答真正需要你的决定，最后检查保存的结果。



产品界面截图

对话始终是主入口，项目状态在侧边保持可见。

01.01

五个操作

1. 点击“新建对话”。确认默认项目文件夹，或选择另一个可写位置。
2. 写出一个具体请求：地点、时间、过程和希望得到的结果。缺少的细节可以稍后决定。
3. 确认 Agent 推荐的 KI。KI 可以已经存在，但对应科学软件尚未在这台 Mac 上验证。
4. 打开“本次运行的数据”。公开或可推导的数据使用“让 Agent 解决”；私有文件由你自行添加。
5. 让 Agent 运行真实模型。查看图表和指标，然后打开项目文件夹检查实际文件。

01.02

如何判断运行是真实的

- 项目显示“模型运行”阶段完成，而不只是生成了一份计划。
- outputs 文件夹中存在由科学模型生成的文件。
- 结果记录了所用执行程序、输入文件、日期、参数和验证检查。
- 图表关联到已保存的数据，而不是没有数据支撑的示意图。

> **建议** 看起来完整的回答并不等于已经运行。保存的输入、命令记录、输出和检查结果才是证据。

完整案例：用 APEX 模拟 50 ha 雨养玉米

本章是整本手册的实操主线。我们用一个真实验收案例：模拟美国得州 Riesel 附近的雨养玉米，时间为 2010–2012 年。你会新建本地项目、选择 APEX KI、让 Agent 准备公开天气与模型输入、处理一次数据选择、运行经过验证的 APEX0806、查看结果图，并检查保存的文件。下方全部是实际 GeoForge 项目的局部截图，不是示意图。

02.01

开始前先明确这次要完成什么

这次的目标是完成一个保存在项目内部的真实验收运行，而不是直接完成可发表的校准研究。主要情景是 Riesel 附近一个合成的 50 ha 雨养玉米地，使用 2010–2012 年 NASA POWER 公开天气，并按 KI 允许使用 20 年预热期。成功以后，项目文件夹中必须留下真实 APEX 输出、检查指标、图表和数据来源记录。

- ✓ 模型：APEX0806；在 macOS 上通过 KI 记录的 Wine 运行方式执行。
- ✓ 输入：由 KI 工具准备天气、站点、土壤假设、作物管理和控制文件。
- ✓ 证据：真实运行完成、产量为有限数值、水量平衡合理、生成图表并保存来源记录。

成功标准

你能够用一句话说清地点、时间、作物、面积以及最后要检查的证据。

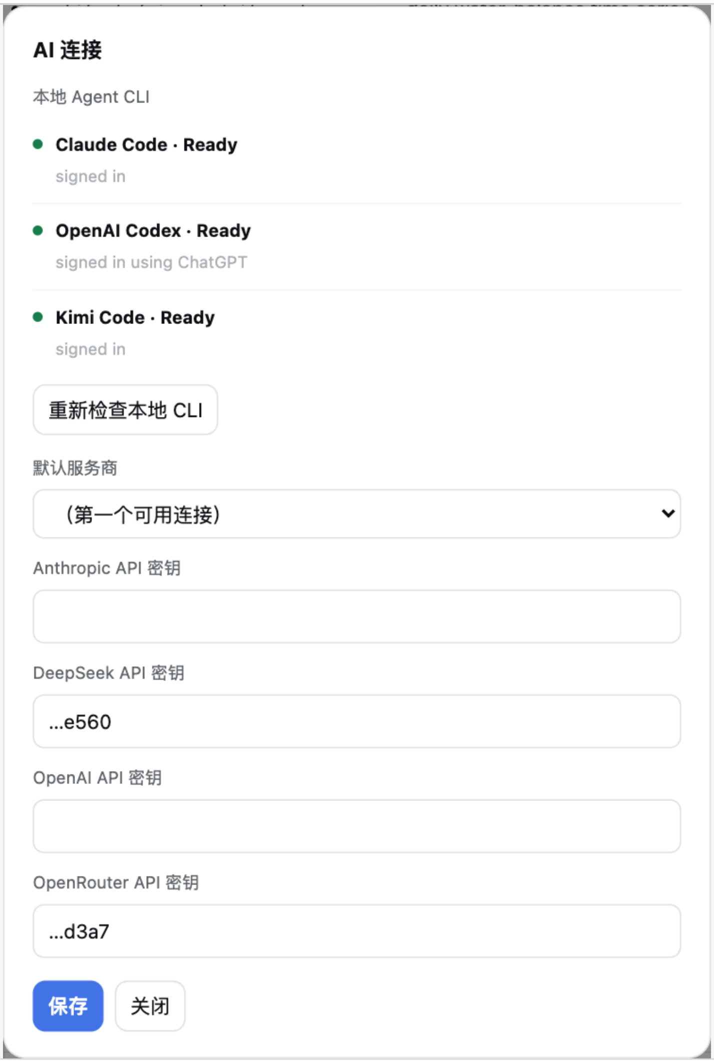
如果没有成功

如果研究问题仍很宽，先选一个地点和较短时间段；第一次真实运行成功以后再扩大范围。

02.02

确认至少一个 AI 连接可以使用

1. 点击左下角的“AI 设置”。
2. 选择一个已就绪的本地 CLI（Claude Code、Codex 或 Kimi），或者配置好的 DeepSeek API。
3. 本地 CLI 在终端登录后点击“重新检查”；API 则保存密钥并完成连接检查。



操作位置 在这里查看每个服务商的状态。“已安装”“已登录”和“可用”是三项不同检查。

成功标准

该服务商可以在对话顶部选择，并且不再显示需要登录或更新的警告。

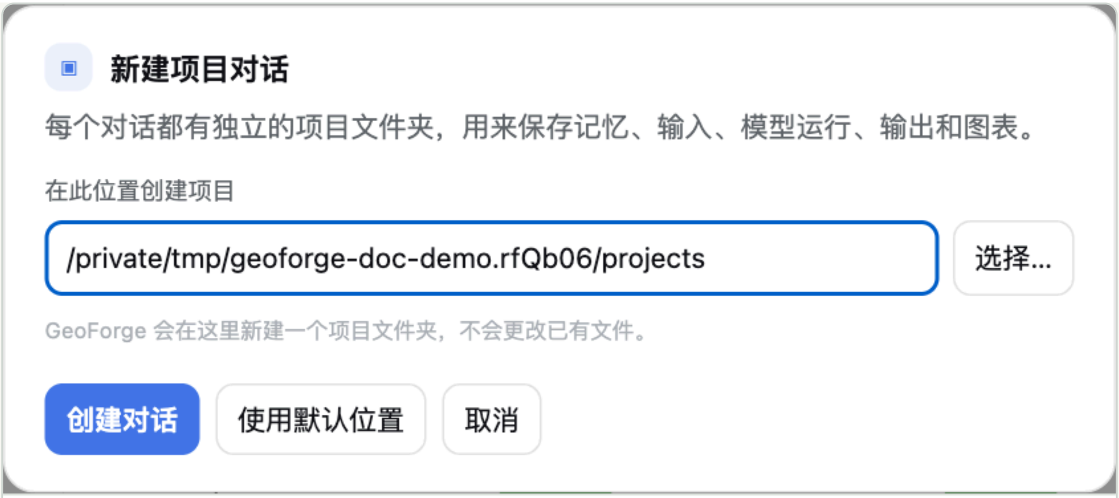
如果没有成功

在终端执行该 CLI 自带的登录命令，返回 GeoForge 后点击“重新检查”。AI 登录失败时，不要重新安装科学模型。

02.03

为这个案例新建一个项目文件夹

1. 点击“新建对话”，GeoForge 会打开“新项目对话”窗口。
2. 使用建议的父文件夹，或选择另一个可写的本地位置。
3. 点击“创建对话”。GeoForge 会新建一个带名称的子文件夹，不会修改父文件夹中已有的文件。



操作位置 对话开始前即可看到保存路径。这个文件夹同时承担持久记忆和模型工作区。

成功标准

左侧出现新的对话，窗口底部显示的路径指向该对话独立的项目文件夹。

如果没有成功

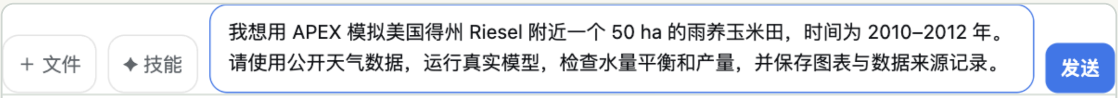
选择用户“文稿”目录下的文件夹。不要使用只读磁盘、应用程序包内部，或正在同步大量文件的云盘目录。

02.04

用普通语言描述模拟任务

你不需要把 APEX 的每个字段都列出来。只要告诉 Agent 科学范围和希望得到的证据，技术字段由 KI 合同补齐。

> **可以直接输入这段话**
>
> 我想用 APEX 模拟美国得州 Riesel 附近一个 50 ha 的雨养玉米田，时间为 2010–2012 年。请使用公开天气数据，运行真实模型，检查水量平衡和产量，并保存图表与数据来源记录。



操作位置 地点、时间、过程和输出足够明确即可开始。真正需要科学决定时，Agent 会稍后再询问。

成功标准

Agent 的第一条回复会先复述情景，并说明准备使用哪个 KI，然后才调用工具。

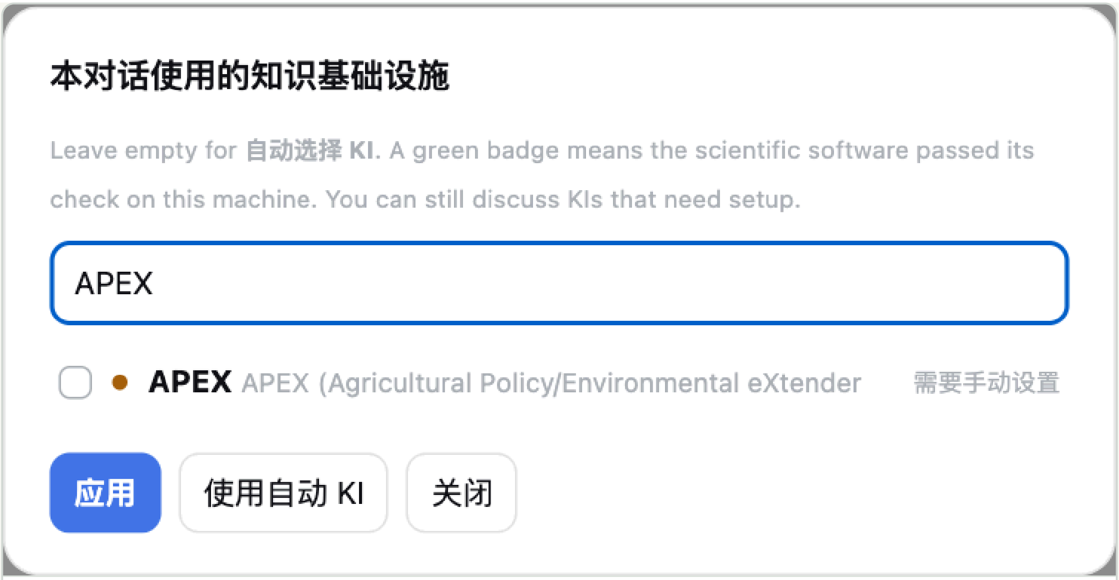
如果没有成功

如果回复太笼统，补充缺少的地点、日期、作物或过程，以及希望得到的结果；不要一次粘贴几十个没有解释的模型参数。

02.05

确认本对话使用 APEX KI

1. 点击对话顶部的 KI 按钮。
2. 搜索 APEX 并选中。只有在确实希望 GeoForge 比较多个模型时才保留“自动 KI”。
3. 点击“应用”。开始运行后，该 KI 会固定在这个对话中。



操作位置　状态徽章表示科学软件是否通过本机检查；这与当前项目的数据是否准备好是两回事。

成功标准

对话顶部的 KI 标记旁显示 APEX。

如果没有成功

打开 KI 库。APEX 已存在但未验证时运行设置 Agent；完全找不到 APEX 时，先导入 KI 包再返回对话。

02.06

运行前先读懂顶部状态栏

- APEX 表示这个对话当前绑定的 KI。
- 验证徽章只表示科学软件在这台 Mac 上的状态。
- “文件夹”会在 Finder 中打开当前项目的真实目录。
- “项目状态”打开动态需求、进度、数据来源、校准与结果面板。
- 右侧显示当前对话使用的 AI 服务商和模型。

操作位置　这些控件分别回答四个问题：使用哪个 KI、软件能否运行、文件在哪里、项目正在做什么。

成功标准

不阅读长对话，也能从顶部确认 KI、AI 服务商、项目文件夹和项目状态入口。

如果没有成功

如果服务商或 KI 选择器已锁定，说明对话已经开始。要改变绑定请新建对话，不要在同一个 CLI 记忆会话中途静默更换服务商。

02.07

打开动态生成的项目数据面板

点击“项目状态”。GeoForge 会读取当前选择的 APEX KI 和项目状态，只把本次运行相关的需求按组展示出来，而不是给所有模型显示同一张固定清单。

操作位置　面板先显示目标与五阶段进度，再显示当前 APEX 数据组及其真实准备状态。

成功标准

目标中出现本次 Riesel APEX 案例，数据部分展示 APEX 特定需求，而不是静态演示清单。

如果没有成功

如果面板为空，先等待 Agent 第一条回复并确认 KI，然后关闭再重新打开“项目状态”。仍然失败时，结合对话错误和项目 runs 文件夹进行诊断。

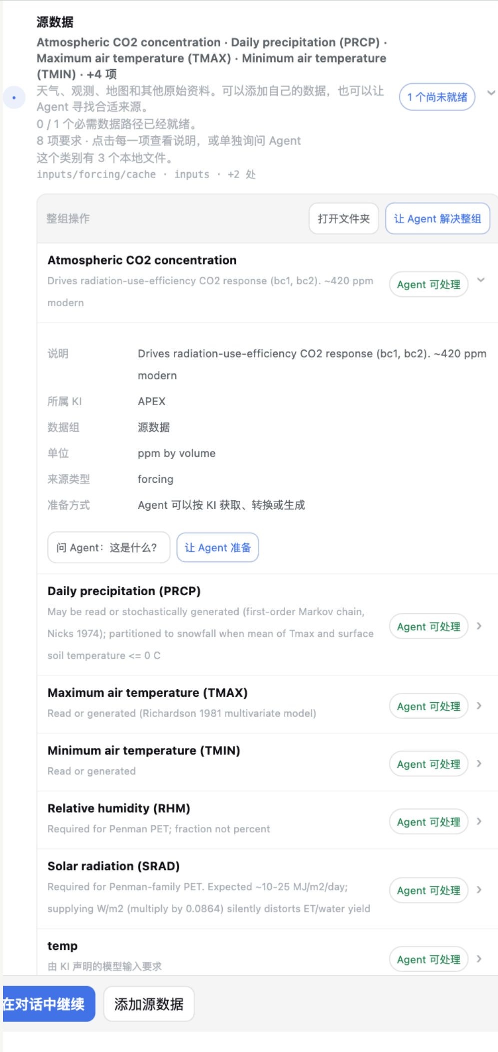
02.08

逐项查看并解决每一个输入要求

1. 找到“源数据”。本案例包括逐日降水、最低与最高温度、大气 CO_2 以及其他 APEX 天气输入。
2. 展开数据类别即可看到当前 KI 声明的全部要求。再展开任意单项，可以查看它的说明、所属 KI、单位、目标文件、格式、来源类型、默认值和有效范围。
3. 点击“问 Agent：这是什么？”可以针对当前项目获得只读解释。Agent 必须区分单项证据和类别级别的就绪状态。

4 单项可以选择“让 Agent 准备 / 查找 / 一起决定”；需要整体处理时再选择“让 Agent 解决整组”。

5 Agent 会先复用并验证已有文件，再按 KI 下载、转换、推导或生成。来源网址、请求参数、校验和、单位和转换步骤都保存在项目中。



操作位置 类别保持简洁，但每一个 KI 要求都可以单独展开、询问 Agent，或独立解决。

成功标准

每个声明项都能独立展开，两个单项按钮都可用，并且“打开文件夹”能进入当前项目中的真实文件。

如果没有成功

私有文件使用“添加源数据”。公开下载受阻时，等待结构化“需要你”请求，不要随意把替代文件复制进模型目录。

02.09

处理一次真正的“需要你”选择

示例项目发现公开站点记录存在缺口。GeoForge 不会把问题埋在一大段 AI 文字中，而是弹出一个简短选择：用

NASA POWER 补齐缺失时段，或者等待你提供本地站点文件。

操作位置 可以选择一项、让 Agent 解释差异，或暂不处理。确认之前不会替换任何文件。

成功标准

选择后，决定会写回对话，Agent 从同一个项目状态继续。

如果没有成功

点击“询问这些选项”。Agent 应用简单语言解释它们对来源记录、一致性和科学解释的影响，然后再次给出选择。

02.10

不离开对话即可添加文件或固定技能

1. 点击“文件”打开系统文件选择器，或把文件直接拖到消息栏。
2. 只有希望在本对话固定某个可复用流程时才点击“技能”；否则 Agent 可以自动选择合适技能。
3. 附件会复制到当前项目的输入区域，同一项目后续消息仍可使用。

＋ 文件 | ✦ 技能 | 补充一个本地观测文件，或在本对话固定一个绘图技能。 | 发送

操作位置 文件、技能、输入框和发送按钮都位于当前对话底部。

成功标准

发送前附件会出现在列表中，或“技能”按钮会显示已固定技能的数量。

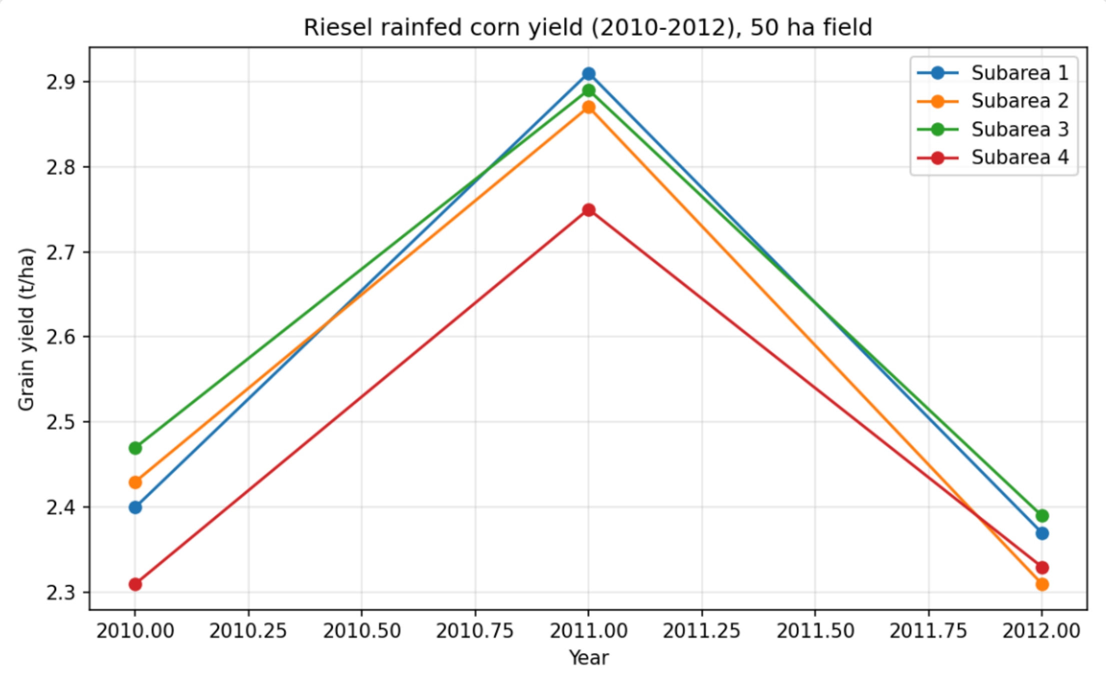
如果没有成功

大文件被拒绝时，可通过 Finder 复制到项目 inputs 文件夹，再告诉 Agent 相对路径；原始文件不要放进模型可执行目录。

02.11

从保存的产量图检查真实运行

完成后的验收案例得到 2010–2012 年有限的玉米产量：范围为 2.31–2.91 t/ha，均值为 2.54 t/ha。Riesel 参考模板保留了多个子区域，因此保存的图中有四条序列，尽管主要配置的子区域为 50 ha。这类细节应当明确检查，而不是隐藏。



Riesel annual corn yield

操作位置 这张图来自 artifacts 下的真实文件，底层年度 CSV 同样保存在项目中。

成功标准

图表能在对话中显示并作为已保存 artifact 打开，其数值与解析后的 CSV 和 Agent 总结一致。

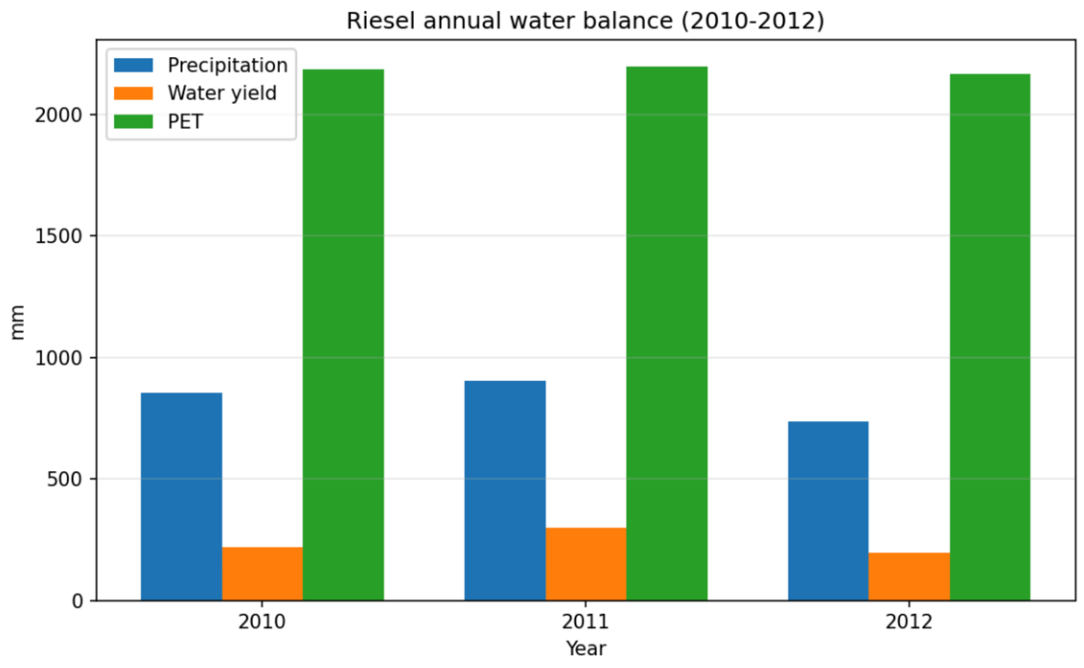
如果没有成功

如果出现图表却没有保存的数据表或模型输出，不要把它当作运行证据。要求 Agent 说明源文件、解析程序和 artifact 的准确路径。

02.12

不要只看产量，还要检查水量平衡

本案例以降水对比径流、蒸散发、深层渗漏、壤中流和储量变化，水量平衡闭合误差为 0.47%。较小的闭合误差说明内部计算一致，但它本身不能证明所有输入和参数在科学上都正确。



Riesel annual water balance

操作位置 这张图必须结合年度水量平衡 CSV 和闭合误差计算一起检查。

成功标准

结果中明确写出闭合计算和值，所有图中数值为有限值，并且源 CSV 已保存。

如果没有成功

先检查单位、日期对齐、预热期排除、缺失值处理以及是否包含储量变化，再考虑修改模型参数。

02.13

打开项目文件夹，逐项检查证据

1. 点击对话顶部的“文件夹”，Finder 会打开这个对话对应的项目目录。
2. 打开 inputs 检查源数据与已准备数据；打开 outputs/APEX 检查可运行工作区和原始模型输出。
3. 打开 artifacts 检查解析后的 CSV 和图表；打开 runs 检查编排脚本、进度和执行记录。
4. 打开来源记录 JSON，确认 NASA POWER 网址与请求参数、校验和、转换步骤、执行程序、日期和模型版本。

本案例应看到的主要目录

```
project/
├── inputs/                  源数据与已准备输入
├── outputs/APEX/            可运行 APEX 工作区与原始输出
├── artifacts/               年度 CSV、产量图、水量平衡图
├── runs/                    编排脚本与执行记录
├── calibration/cases/       已保存的校准计划
├── memory/                  持久项目记忆
└── session.json             对话与服务商绑定信息
```

成功标准

另一位研究人员不依赖对话文字，也能找到准确输入、执行程序、命令或工作流、原始输出、解析表格、图表和来源记录。

如果没有成功

如果对话声称文件存在，但 Finder 中找不到，要求 Agent 给出项目绝对路径和 artifact 相对路径。以文件系统为事实依据。

02.14

理解“已保存校准计划”到底意味着什么

本次运行在 calibration/cases 下保存了一份校准案例计划，其中包括产量和水量平衡等目标、候选参数、优化策略与留出验证设计。由于 APEX 适配器仍标记为需要编写，校准引擎并未实际运行。因此 GeoForge 正确地把它称为“计划”，而不是“已校准”或“可推广”的模型。

- ✓ 计划已保存：以后可以继续审核和完善这个工作流。
- ✓ 实际校准：需要可工作的项目专属适配器和真实观测数据。
- ✓ 已校准：需要真实模型评估成功并得到目标函数结果。
- ✓ 已验证或可推广：还必须通过独立留出验证。

成功标准

项目中存在可读的案例计划；除非同时存在引擎运行记录和留出指标，否则结果不会声称已完成校准。

如果没有成功

添加真实观测后，在“项目状态”的校准区域让 Agent 完成项目本地适配器。不要为了让单个项目通过而修改共享引擎或原始 KI。

说明 诚实地说明停止位置也是可复现性的一部分。保存的计划有用，但它不是校准结果。

在 macOS 上安装与启动

GeoForge 是本地桌面应用。应用包包含界面和共享运行框架；具体科学模型仍可能需要安装、许可证或单独下载。

03.01 安装应用

1. 下载与 Mac 匹配的版本。Apple 芯片使用 arm64；较早的 Intel Mac 在提供时使用 x86_64。
2. 解压后把 GeoForge Desktop.app 移到“应用程序”。
3. 首次启动时 macOS 可能要求确认。只有在确认下载来源可信后，才在“系统设置 › 隐私与安全性”中允许打开。

03.02 选择工作保存位置

每个新对话都会在 GeoForge projects 目录中建议一个文件夹。你可以接受，也可以选择其他本地位置。项目应放在应用程序包之外，这样更新应用不会覆盖工作。

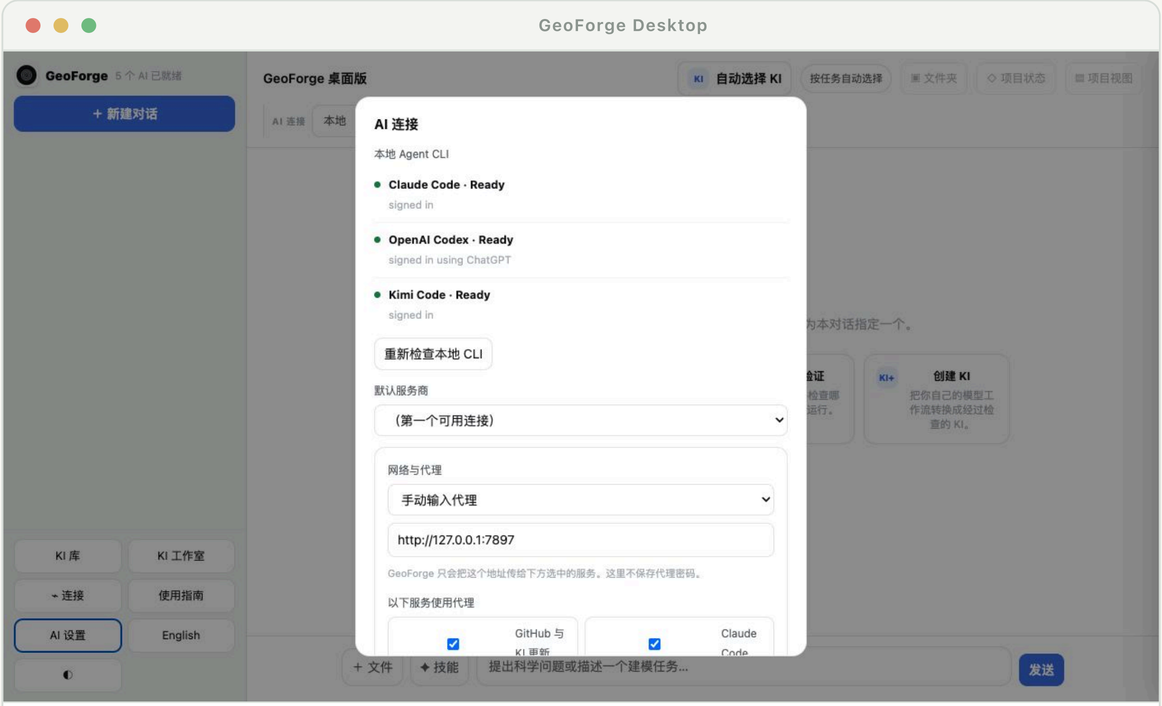
> **说明** 大型模型应选择空间充足的磁盘。模型正在大量写文件时，尽量避免使用云同步目录。

04

CLAUDE CODE、CODEX、KIMI 与 DEEPSEEK

连接 AI 服务

GeoForge 可以使用已登录的本地 CLI，也可以使用 API。两者都通过同一套 KI 运行框架和项目工具；服务商提供推理与语言能力。



产品界面截图

AI 设置集中显示 CLI 就绪状态、默认服务商、网络路由、Kimi 权限、API 密钥和完整链路测试。

04.01

本地 CLI 模式

- ✓ Claude Code：在终端安装并登录，然后在 GeoForge 中点击“重新检查”。
- ✓ OpenAI Codex：安装或更新 CLI，并完成正常登录流程。
- ✓ Kimi：安装 Kimi Code、完成登录，并从本地服务列表中选择。
- ✓ GeoForge 使用同一用户账户启动服务，但把工作目录限制在当前项目中。

04.02

API 模式

使用 DeepSeek 或其他受支持 API 时，在 AI 设置中填写密钥，或通过支持的环境变量提供。系统会发送一次真实的单消息探测，确认密钥、端点、模型和网络都能协同工作。

> **注意**　“已安装”不等于“已登录”，“已登录”也不等于所选模型可用。GeoForge 会分别显示这些状态。

04.03

只让需要代理的服务走代理

1. 打开“AI 设置”，选择“使用 Mac 系统代理”“手动输入代理”或“不使用代理”。
2. 手动代理可填写本地地址，例如 http://127.0.0.1:7897。GeoForge 不保存代理凭据；需要认证时请使用本地已认证代理。
3. 分别为“GitHub 与 KI 更新”、Claude、Codex、Kimi 或 API 服务启用代理。不能因为 Claude 需要代理，就让 Kimi 和 DeepSeek 被动使用同一路由。
4. 所选服务商的 Agent 在运行 Git、pip、curl 和下载命令时会使用同一路由。开始长时间安装前，先点击“测试 AI 与 GitHub”。

成功标准

服务商探测与 GitHub 探测都通过预期路由返回真实成功结果。

如果没有成功

如果以前可以直连、现在失败，请对照 Mac 系统代理检查所选模式，并测试代理地址本身。DNS 错误、连接超时、登录失败和模型不可用是不同问题。

04.04

选择 Kimi Code 文件访问范围

- ✓ 推荐“项目范围”。Kimi 可以访问当前项目、所选 KI、已批准的数据路径及自身登录/会话状态；其他个人文件仍被阻止。
- ✓ 当 Kimi 需要用户技能目录等外部文件夹时，GeoForge 会弹出范围明确的权限窗口。请选择“仅允许一次”或“此项目始终允许”，不要开放整个主目录。
- ✓ “完整电脑访问”是兼容性后备方案。只有项目范围模式无法工作并且已理解风险时才使用。

> **说明** 新的未授权路径应只弹出一次权限请求。如果选择“此项目始终允许”后仍反复出现，应检查项目保存的权限策略，而不是不断重复批准。

04.05

哪些内容会发送给服务商

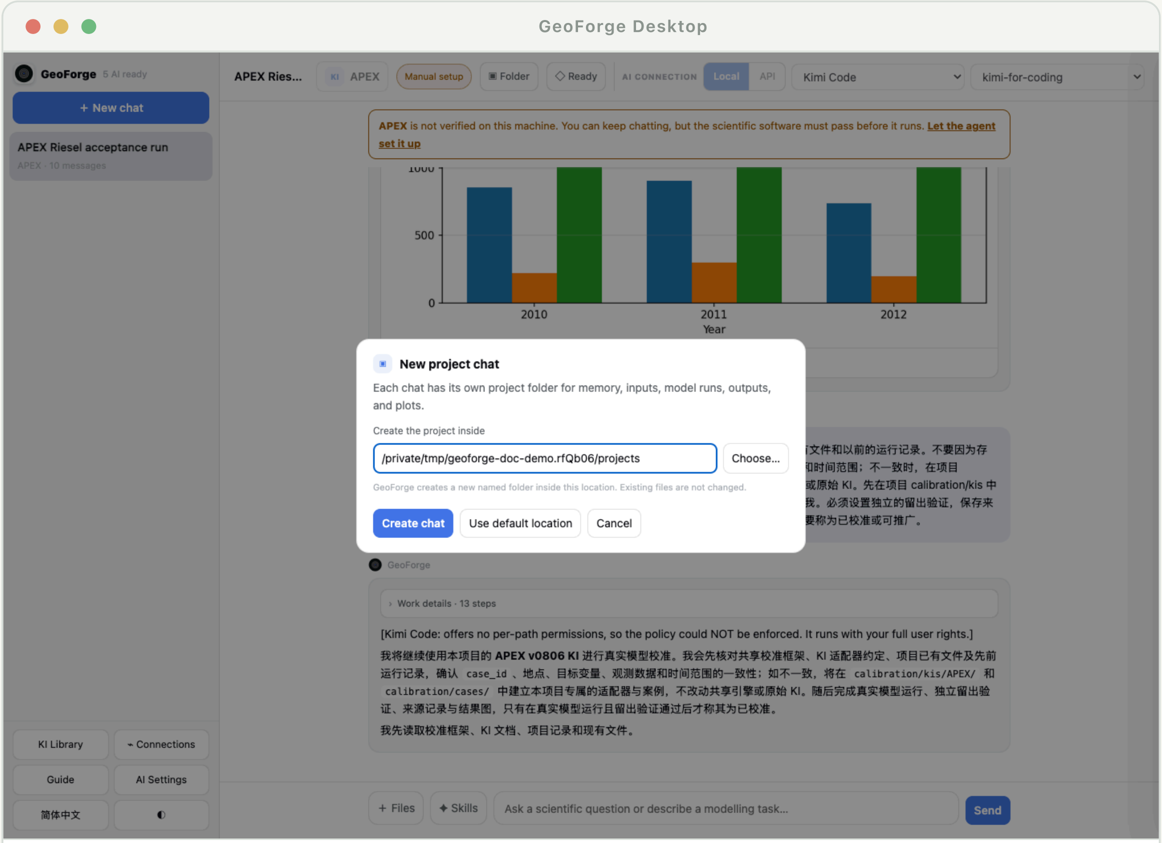
对话、相关 KI 指令以及所选工具结果可能发送给当前 AI 服务商。本地模型文件不会自动完整上传；Agent 只读取任务工具明确提供的内容。处理机密数据前，请先查看服务商条款。

05

每个对话都有一个持久的本地文件夹

项目、对话与记忆

GeoForge 不依赖服务商永久记住你的项目。每个对话都有本地项目文件夹和持久记录。服务商支持时会保存其原生会话 ID，但本地项目状态始终是事实依据。



产品界面截图

开始对话前可以看到并修改默认路径。

05.01

项目文件夹包含什么

- ✓ 用于继续工作的对话历史和精简项目状态。
- ✓ 所选 KI、执行策略、服务商/会话元数据和进度事件。
- ✓ 源文件、已准备输入、模型工作区、输出、图表和报告。
- ✓ 来源记录：数据从哪里来、如何转换、哪个命令生成了结果。

05.02

任务运行时切换对话

正在运行的对话会显示活动状态，并在自己的任务中继续。你可以新建或打开另一个对话，不会让整个应用看起来“卡死”。返回原项目时会看到最新事件、待处理请求或结果。

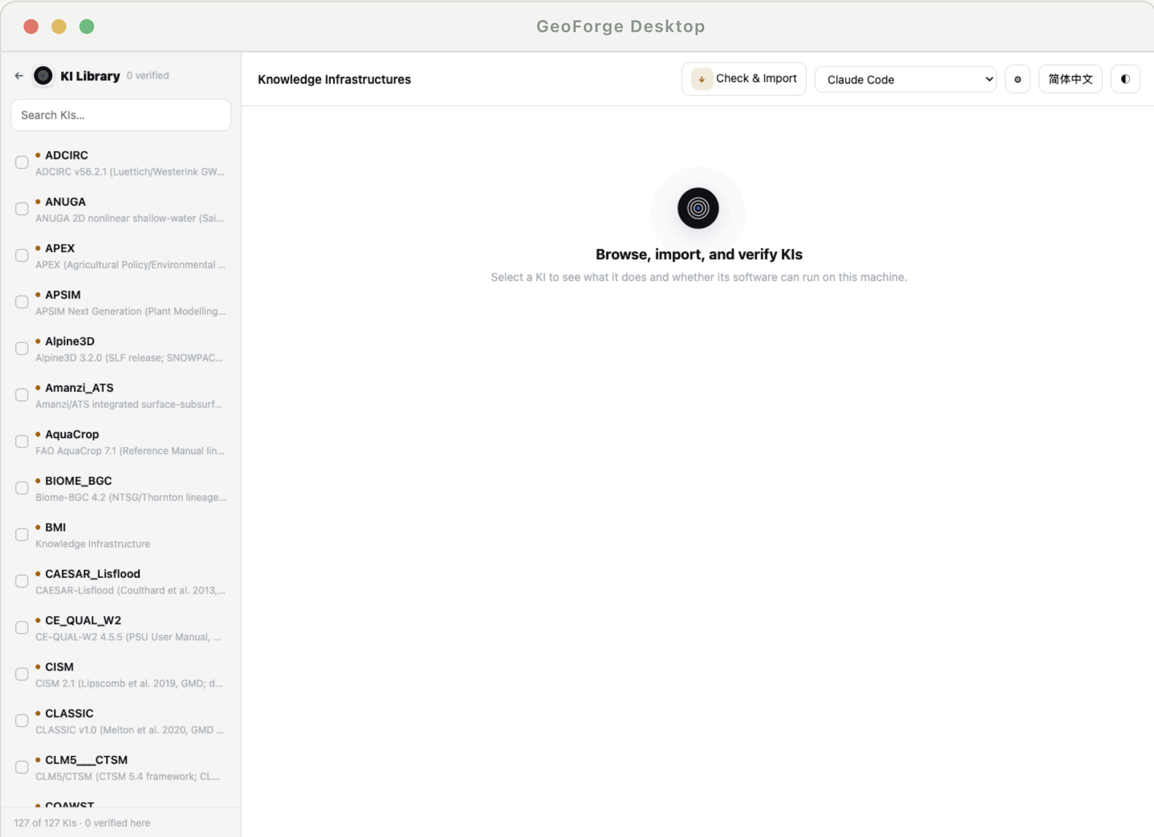
> **建议** 一个对话对应一个科学目标。当科学问题、研究区域或模型实验发生实质变化时，建议新建项目。

06

可用性、验证、导入与可信状态

了解 KI 库

Knowledge Infrastructure（KI）是 Agent 使用的模型专属操作知识。它可包含安装方法、工具、诊断、数据契约、示例、文档摘要和结果解释规则。



产品界面截图

KI 可以已经存在，但底层模型尚未在这台 Mac 上验证。

06.01

不要混淆三种状态

- ✓ KI 可用：GeoForge 可以读取模型指导与工具。
- ✓ 软件已在本机验证：所需执行程序已在本机通过 KI 的真实预检。
- ✓ 项目已就绪：当前情景已经具备足够的有效输入和决策，可以运行。

06.02

导入并验证 KI

1. 点击“导入 KI”，选择 KI 包或仓库文件夹。
2. 运行验证器。导入前会检查结构、声明的工具、路径和安全规则。
3. 查看科学软件名称、版本、许可证、来源和最近检查结果。
4. 如果 KI 已存在但模型软件尚未就绪，使用“设置并验证”。

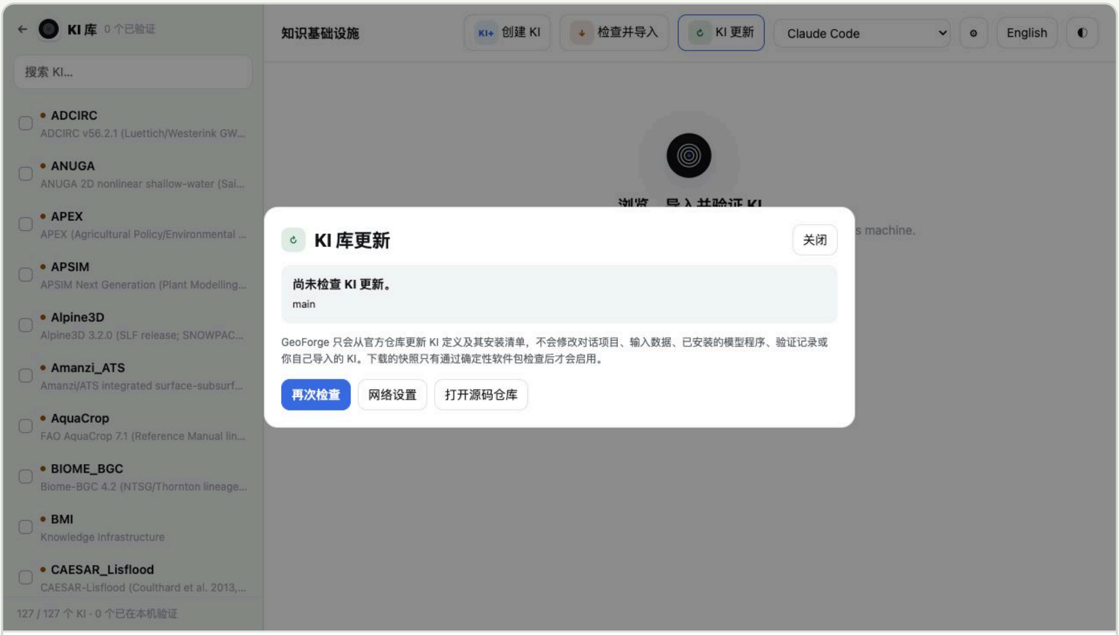
> **说明**　验证是本地且针对具体版本的。Linux 服务器上验证过的模型，不会自动在 macOS 上视为已验证。

06.03

从官方 main 分支自动更新 KI

GeoForge 启动时会检查官方仓库 main 分支中的模型 KI 定义和安装清单。下载快照只有通过确定性软件包检查后才会启用；网络或验证失败时，会继续使用上一次通过验证的快照。

- ✓ 更新器可以新增、修改或删除官方 KI 文件，并报告分支、时间、网络路由和变更摘要。
- ✓ 它不会覆盖对话项目、输入数据、已安装模型二进制、验证历史或用户自行导入的 KI。
- ✓ 无法直接访问仓库时，启用“GitHub 与 KI 更新”代理开关。



操作位置　更新报告会说明来源分支，以及更新器被允许修改的内容范围。

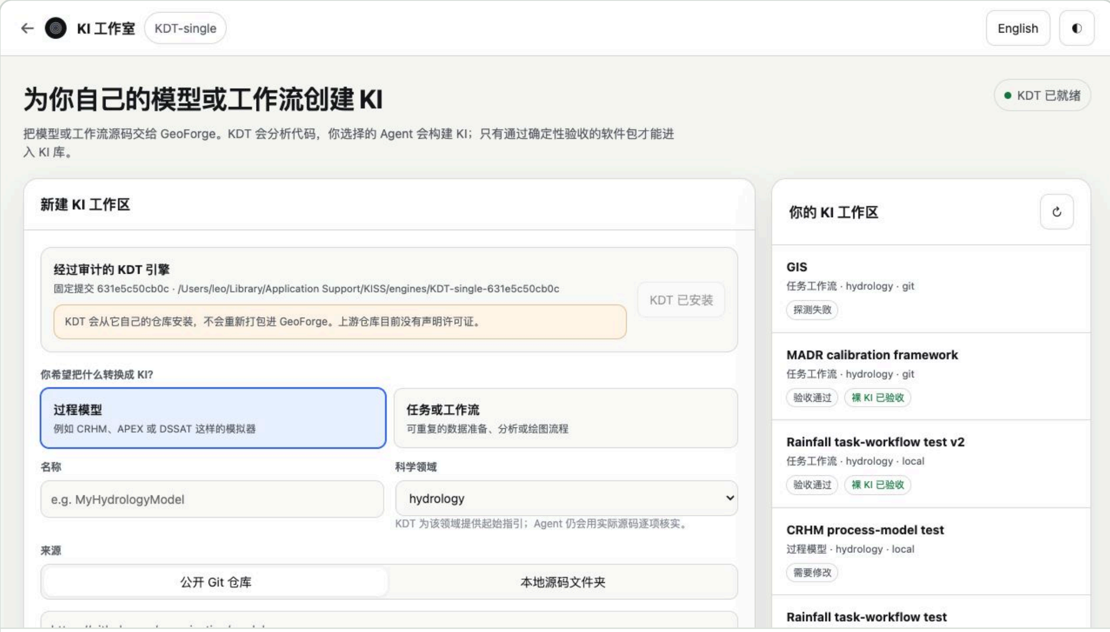
06.04

在 KI Studio 中创建 KI

KI Studio 使用经过审计的 KDT-single 引擎分析公开 Git 仓库或本地源码文件夹。它既可处理过程模型，也可处理可重复的任务/工作流。可以选择已有科学领域，也可以自行输入；没有 KDT 先验指导的领域必须完全根据源码核实。

1. 创建独立工作区并运行 KDT 源码探测。
2. 为构建阶段选择 AI 服务商；后续修复轮次也可以更换服务商。

3. 只有请求面板明确提示缺少受保护论文、文档、凭据或许可资产时，才由用户提供。
4. 先运行 KDT 门槛，再运行 GeoForge Desktop 软件包/导入门槛。模型安装与真实预检要在目标电脑上稍后完成。



操作位置 三个结果层级可避免把结构通过的裸 KI 误认为已经安装完成的科学软件。

注意 “裸 KI 通过 KDT”“可导入 Desktop”“科学软件已验证”是三个独立结果。前两项都不能证明科学执行程序已经可以运行。

如何提问，以及 AGENT 会决定什么

通过对话完成工作

用户不需要把科学目标翻译成几十个文件名。先提出问题，Agent 会利用 KI 找到技术要求，只把无法安全推断的决定交给你。

07.01

一个好的首条消息

尽量包含过程、地点、时间、尺度以及希望比较或输出的内容。例如：“模拟新乡附近 2000–2020 年雨养玉米产量，并比较增温情景与基准情景。”没有决定的内容可以省略。

07.02

Agent 应自动处理什么

- 选择或推荐合适 KI，并简要说明理由。
- 检查软件就绪状态，并在模型工作区内修复常规安装问题。
- 把 KI 数据契约转成简短的项目数据计划。
- 在允许范围内下载公开数据、转换格式、生成可推导参数并验证输入。
- 运行模型、检查输出、绘图并解释不确定性，不隐藏失败。

07.03

GeoForge 何时应询问你

- 存在多个合理科学选项，而且会改变结果解释。
- 需要私有文件、登录、许可证、受保护下载或 macOS 权限。
- 操作会写入项目外部、安装系统软件或访问敏感位置。

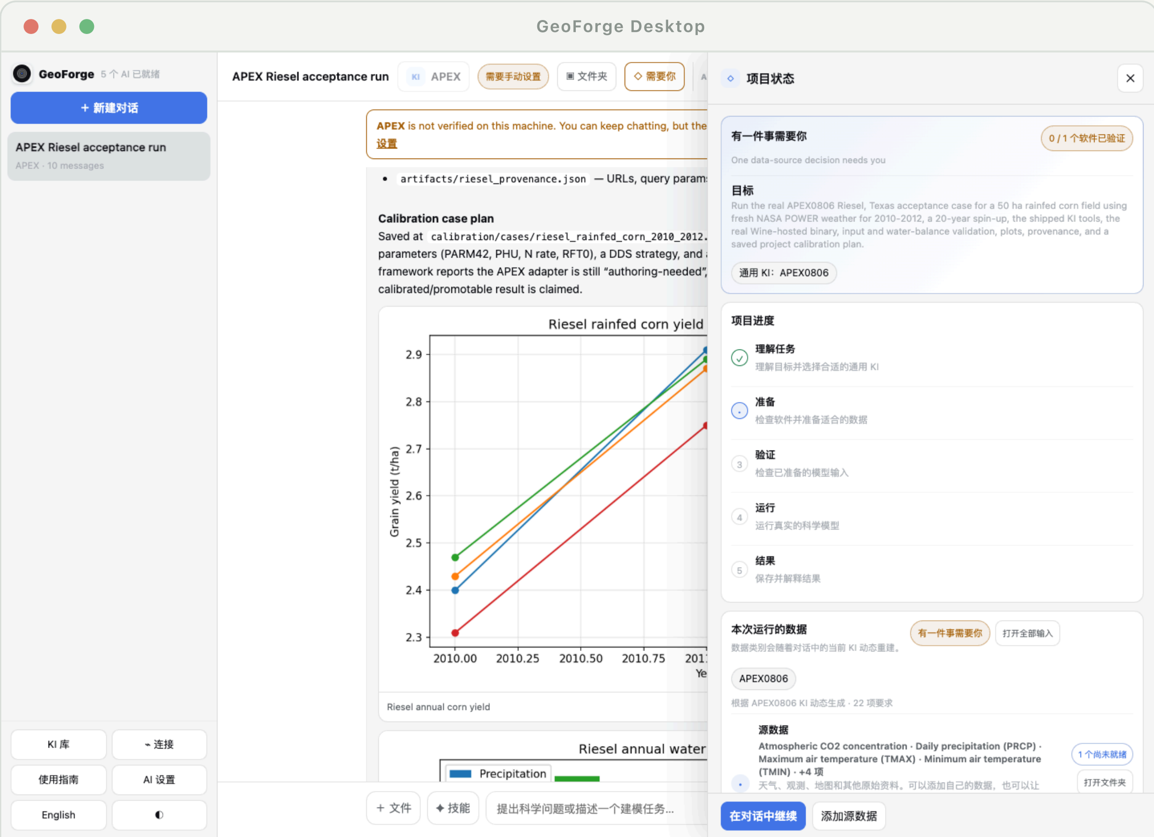
> **建议**　　你随时可以问：“哪些是你的假设，哪些是实测数据，哪些是生成的数据？”

08

根据 KI 和当前运行动态显示所需数据

准备项目数据

Agent 理解情景并选择一个或多个 KI 后，数据面板会动态生成。它不是固定清单，也不会把每个模型内部文件都变成用户的工作。



产品界面截图

概览保持简洁；技术要求仍可在详细信息中查看。

08.01

四类简洁数据状态

- 源数据：天气、观测、地图、实测数据或其他原始资料。
- 空间与网格参数：由源数据构建的流域、图层、网格、HRU、河段或点位。
- 模型选择与校准：参数、目标函数、范围和有依据的默认值。
- 初始与边界条件：模拟开始时的状态，以及模型区域周边条件。

08.02

使用“让 Agent 解决”

1. 展开类别，查看“已就绪”“处理中”“待审核”或“需要你”等状态。
2. 点击“让 Agent 解决”。Agent 会读取所选 KI 的数据契约和当前情景。
3. 它会寻找允许使用的公开来源、转换格式、推导技术文件并记录来源。
4. 如果需要真实决定或权限，任务会进入“需要你”，并显示具体请求。

08.03

查看实际文件

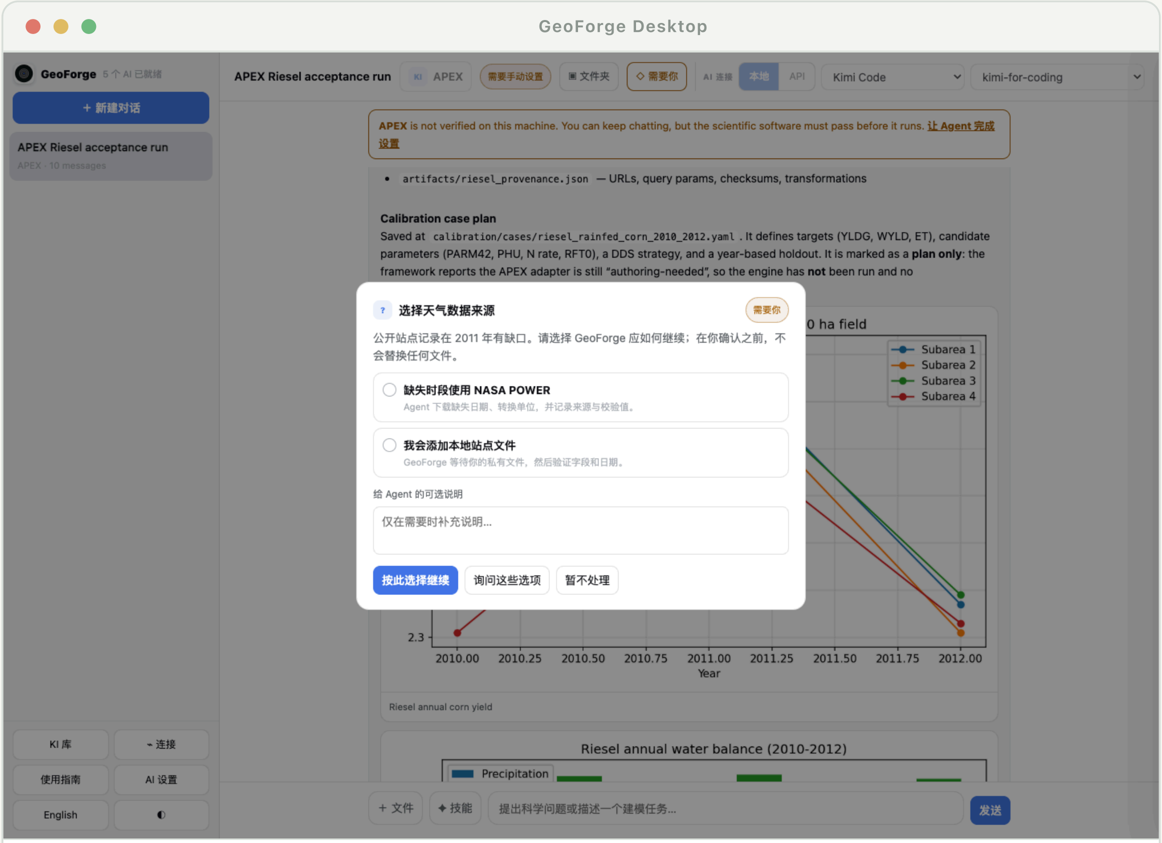
使用面板中的“打开文件夹”或对话顶部的“文件夹”按钮。GeoForge 会在 Finder 中打开项目位置，你可以直接检查源数据、准备后的输入和输出。详细条目应显示本地路径、来源网址或引用、转换过程、验证、日期，以及可用时的校验值。

> **注意**　“已就绪”表示该要求已有通过验证的项目路径，并不表示每个数值都是实测或没有不确定性。

决定、权限、许可证与手动下载

处理“需要你”

当 Agent 无法安全地独立继续时，GeoForge 应显示结构化请求，而不是把命令埋在长消息里。请求会说明阻塞内容、原因，以及恢复工作所需的最小操作。



产品界面截图

可以选择安全操作、拒绝操作，或回到对话讨论其他方案。

09.01

常见请求类型

- 权限：允许一个文件、文件夹、执行程序或范围明确的命令。
- 下载：打开需要接受条款、验证码或账户的数据网站。
- 许可证：确认你有权安装或运行受限制软件。
- 科学决定：在 Agent 说明影响的多个方案中做出选择。
- 私有输入：选择本地文件，不会把它公开，也不会悄悄替换为合成数据。

09.02

动态权限策略

GeoForge 可以在项目中授予范围明确、可恢复的动态权限。合适的选项包括“仅允许一次”“此项目始终允许”“选择其他文件”和“拒绝”。不应默认允许访问整个主目录或无限制执行终端命令。

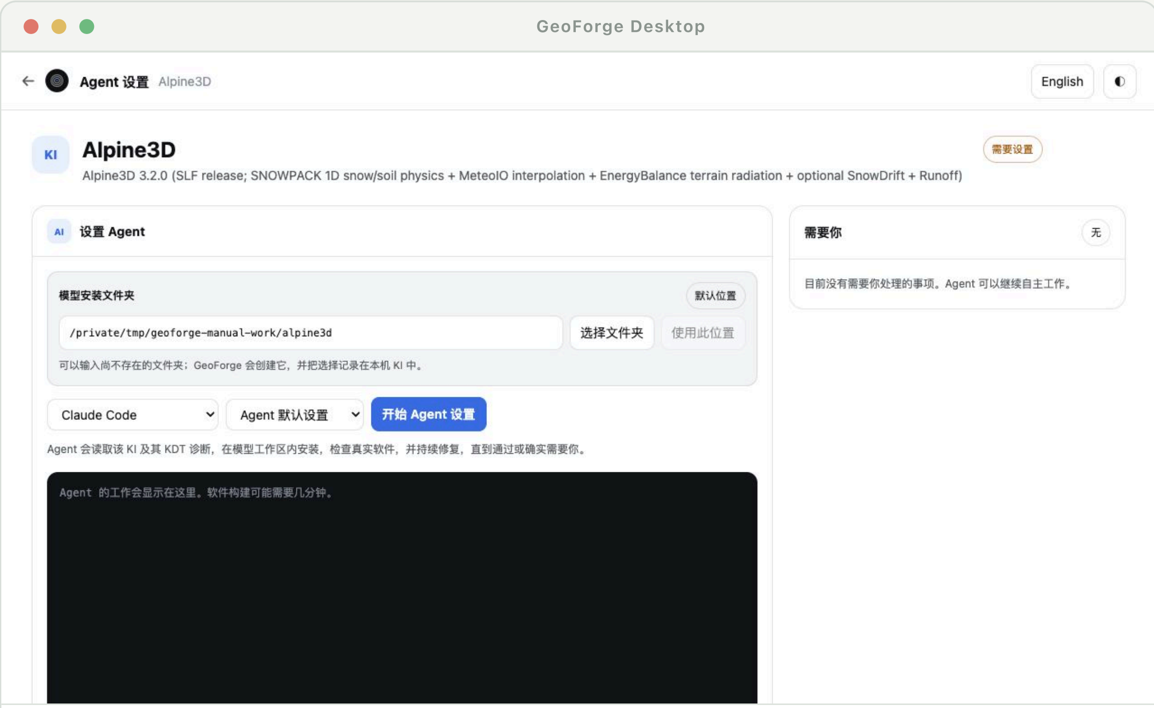
> **建议** 如果选项不清楚，选择“在对话中讨论”。Agent 应解释风险，并尽可能找到项目内的替代方案。

10

由 AGENT 主导安装，并在关键处询问你

设置并验证模型软件

设置 Agent 会在模型工作区中执行 KI 的 KDT 配方：准备工具和环境，获取或编译正确版本，并运行真实预检。普通失败应自动诊断和重试。



产品界面截图

启动设置 Agent 前选择安装目录；尚不存在的路径可由 GeoForge 自动创建。

10.01

选择并记录模型安装目录

1. 接受“默认位置”、选择现有文件夹，或直接输入新路径。文件夹可以尚不存在；GeoForge 验证后会自动创建。
2. 为模型使用稳定的安装工作区，不要把软件安装在临时对话目录中。
3. GeoForge 会把所选位置写入安装配置和本地 KI 安装记录，后续对话与预检会解析到同一路径。
4. 可移植 KI 占位符会在运行时解析。KI 不应依赖开发者硬编码的 /Users/... 路径或 Windows 盘符。

> **建议**　修改输入框不会自动搬迁已有安装。请选择已经记录的安装位置，或执行明确的迁移流程。

10.02

验证能证明什么

1. KI 已准备完成，所需共享工具可以导入。
2. 所需运行环境和系统依赖已经存在。
3. 目标软件版本已针对本平台获取或编译。
4. KI 的真实预检会启动执行程序并检查预期行为。

10.03

Agent 会自行修复什么

- ✓ Git 分支或标签不匹配、可重试的克隆失败、构建目录或项目内路径问题。
- ✓ 缺少项目内 Python 包，例如 ki_tools_common。
- ✓ 模型二进制需要复制或链接到受控工作区。
- ✓ KI 已记录修复方法的预检失败。
- ✓ 设置循环没有人为步骤上限。只要修复仍然合理，它会继续执行，直到验证成功、确实需要用户操作，或服务商停止。

10.04

读懂实时运行状态

- ✓ 已连接：服务商进程仍在运行，并且 GeoForge 最近收到过事件。
- ✓ 处理中：可见阶段和步骤数持续变化；大型编译运行数分钟是正常的。
- ✓ 等待中：服务商进程仍在，但正在等待响应、权限、下载或模型命令。
- ✓ 可能卡住：超过提示阈值仍无心跳或新事件。GeoForge 应提供诊断或取消操作，而不是假装仍在推进。

> **说明**　仅凭已用时间不能判断卡死。应结合心跳、最后事件、进程退出状态和服务商错误判断。

10.05

哪些仍需要你

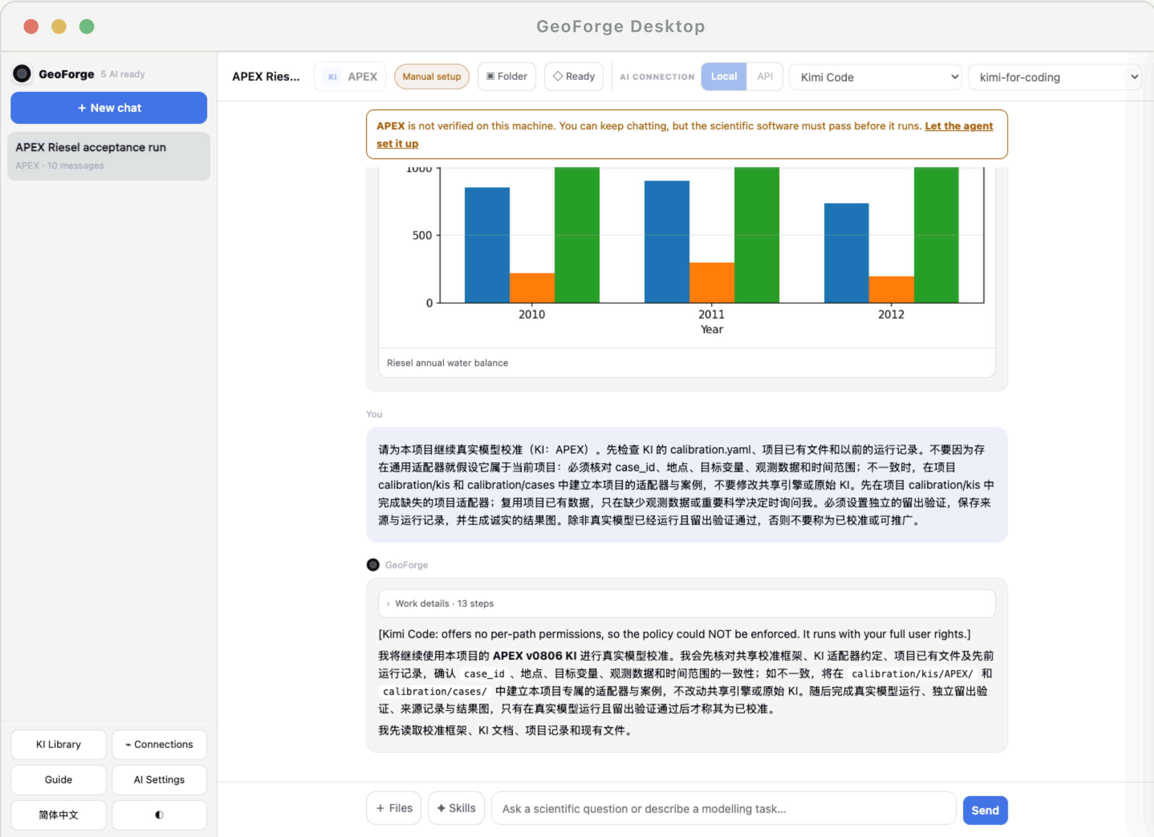
遇到许可证、受保护下载、登录信息、macOS 系统权限或破坏性/系统级修改时，Agent 必须停下来询问你。软件验证与项目数据应分开：缺少天气或 DEM 数据，不应把已成功安装的模型标成损坏。

> **注意**　安装日志只显示“Load failed”是不够的。界面应保留真实的服务商错误、退出状态和最后成功步骤。

从准备好的输入到可信结果

运行、验证与解释

一次成功的 GeoForge 运行包含三层：执行证据、验证证据和科学解释。Agent 应让三层都可见，并保存在项目中。



产品界面截图

结果始终关联到项目文件和验证检查。

11.01

执行之前

- ✓ 确认研究范围、日期、模型版本、单位和情景假设。
- ✓ 验证必需路径、文件格式、缺失值、范围、坐标系统和时间覆盖。
- ✓ 固定或记录准备后的输入，以便复现运行。

11.02

执行之后

- 检查进程退出状态和模型专属完成标志。
- 拒绝空文件、旧文件、截断文件、不可能值或非有限值输出。
- 执行 KI 声明的质量平衡、基准、范围、连续性或参考案例检查。
- 区分模拟结果、观测数据、假设和校准目标。

11.03

批判性阅读结果

模型结果取决于所选结构、数据、参数化和情景。应询问敏感性、留出期表现、缺失过程，以及哪些因素最可能改变结果。运行不完整时，GeoForge 应直接报告失败或限制，而不是围绕它构造科学故事。

图表、表格与项目文件

Agent 可以在对话中直接渲染图表和紧凑表格。每个重要图表也应保存到项目中，以便重新打开、再生成或用于报告。

12.01

对话内可视化

- 图表应标明变量、单位、日期、空间范围，以及数值是观测还是模拟。
- 校准图应区分校准期和验证期。
- 地图应标明投影、分辨率、范围和掩膜规则。
- 可视化错误不应删除底层模型输出；Agent 可以修复后重新渲染。

12.02

打开动态项目视图

“项目视图”根据每个对话的安全项目清单动态渲染。Agent 与所选技能可以发布图表、表格、地图、时间控件，以及二维洪水过程或 BIM 场景等模型专属视图。未来专门化 KI 也可以定义自己的界面，同时把底层结果文件保存在项目中。

- 它不是所有模型共用的一张静态仪表盘。
- 只渲染已声明、位于项目中的结果和受支持组件；不会执行 Agent 任意生成的 HTML 或 JavaScript。
- 使用项目文件夹检查每个可视化背后的源数据，并在以后重新生成。

12.03

添加自己的文件

把文件拖到消息输入框，或点击“+ 文件”打开文件选择器。GeoForge 会按所选策略复制或引用到项目中，记录原始名称，并告诉 Agent 文件位置。大型文件夹应通过文件夹选择器添加，而不是粘贴到对话中。

> **建议**　文本使用消息上的“复制”，项目文件使用“打开文件夹”。复制显示的路径不等于打开或授权访问该路径。

构建可重复使用的校准工作流

校准不是一次性对话。GeoForge 会把它变成保存的工作流：共享校准引擎、模型专属 KI 适配器，以及修改范围、观测或预算后可重新运行的项目案例。

13.01

三个层次

- 校准引擎：跨模型共享的算法、调度、检查点、指标和结果存储。
- KI 适配器：如何写入参数、启动模型，以及如何把输出转换为目标函数值。
- 项目案例：本研究的参数、范围、观测、预热期、校准/验证期、目标函数和计算预算。

13.02

创建工作流

1. 基线运行成功且观测目标通过验证后，再要求进行校准。
2. 查看推荐参数、范围、约束、目标指标和留出验证设计。
3. 在项目 calibration 文件夹中生成适配器和案例。
4. 先运行冒烟测试，再运行所选预算。失败候选仍保留记录，不会消失。
5. 保存最优参数集、完整历史、诊断结果，以及重新运行的命令或按钮。

13.03

以后更新

打开保存的校准案例，调整允许修改的迭代预算、参数范围或目标文件，然后点击“运行校准”。GeoForge 应创建新的运行记录，而不是覆盖旧结果。模型或适配器版本发生变化时，需要重新验证。

注意　校准可以提高拟合度，却不一定更接近真实。始终保留独立数据或验证期，并同时报告两种表现。

技能与 MCP 连接

KI 解释如何操作科学模型。技能提供绘图、统计分析、地理处理或文献工作流等可复用方法。MCP 连接则提供对 GitHub 等外部服务的受控访问。

14.01

在对话中使用技能

点击消息框旁的“技能”，查看或固定本对话的技能。也可以输入斜杠命令，或自然地说“生成发表质量的径流过程图”。Agent 可以自动使用相关已安装技能，而选中列表会明确表达你希望采用的方法。

14.02

连接 MCP 服务

1. 打开“连接”，选择可用的 MCP 服务。
2. 授权前先查看它提供哪些资源和操作。
3. 在服务商可信流程中完成登录，然后返回 GeoForge。
4. 只授权项目所需的仓库、文件夹或权限范围。

> **说明** 连接会扩大 Agent 可访问的内容，但不会替代 KI 验证或项目来源记录。

一致的英文或简体中文体验

语言、复制与键盘操作

界面语言控制标签与指南。对话会跟随你最新消息的语言，因此中文提问应得到中文回答，除非你另有要求。

15.01

语言行为

- 使用应用中的 English/中文 切换导航、状态、弹窗和本指南。
- 科学名称、模型文件名、命令、单位和路径不会被错误翻译。
- 可以要求“answer in English”或“请全程使用中文”来覆盖自动语言判断。

15.02

复制、粘贴与快捷键

- 正常选择文字，并在输入框中使用 Command-C / Command-V。
- 点击助手消息上的“复制”，复制渲染后的文字，不包含内部工具标记。
- 界面提示时可用 Command-K 新建任务；Enter 发送，Shift-Enter 换行。
- Agent 运行时，导航和“新建对话”仍可使用。

隐私与可复现性

GeoForge 以本地为主，但所选 AI 服务商和已连接服务可能接收任务上下文。可复现性也不只是保存输出：项目必须保存生成结果的准确输入和决定。

16.01

使用敏感数据之前

- 检查当前服务商及其数据政策，并确认本地 CLI 是否仍会把提示发送到云端。
- 从对话和普通项目文件中移除密钥、个人身份信息、未公开受限数据和登录凭证。
- 使用项目范围权限，不再需要外部服务时及时断开连接。
- 除非许可条款允许，否则不要上传受版权保护的论文或模型二进制文件。

16.02

最小可复现记录

- GeoForge、KI、模型、适配器和校准引擎版本。
- 源数据标识、网址/引用、访问日期、校验值和许可证。
- 所有转换、单位、投影、缺失数据决定和生成参数。
- 执行命令、环境、随机种子、运行日期、失败记录和验证检查。
- 图表和表格关联到它们所汇总的准确结果文件。

建议 当结果用于论文、报告或决策时，应归档或导出整个项目。

故障排查与术语表

先查看可见状态和项目事件日志。不要在未阅读最后真实错误的情况下反复重启任务：它可能在等待登录、权限、下载或科学决定。

17.01

常见问题

- CLI 显示需要登录：在终端运行服务商自身状态命令，确认是同一 macOS 用户，然后点击"重新检查"。不要把过期探测当成事实。
- 应用看起来卡住：检查活动标记和项目事件。API 调用与模型运行应显示阶段、已用时间，以及"取消/后台继续"选项。
- 找不到 ki_tools_common：把共享工具重新部署到项目环境中，验证工具路径后再重试。
- 预检因缺少数据失败：区分软件预检和项目输入验证。先验证软件，再准备情景数据。
- Git 克隆失败：设置 Agent 应检查标签/分支、网络状态和残留目录，然后安全重试。只有需要登录或受保护访问时才询问你。
- 复制粘贴无效：点击输入框内部，使用"编辑"菜单或 Command-C/Command-V；如果仍无法选择，请报告具体字段。

17.02

术语表

- KI — Knowledge Infrastructure：Agent 使用的模型专属知识、工具、检查和数据契约。
- KDT — 让 KI 对应软件在本机可用的可执行安装与诊断配方。
- 预检 — 完整项目运行前进行的真实检查，用于证明目标软件可以执行。
- 项目运行 — 一次可追踪的执行，包含固定输入、配置、输出和验证。
- 适配器 — 共享框架与某个模型的文件、命令和输出之间的专属桥梁。
- 留出数据 — 不参与校准、用于独立评价的观测或时间段。
- 来源记录 — 数据、参数、代码和结果的来源及转换历史。

17.03

提交有效的问题报告

请包含 GeoForge 版本、macOS 版本和芯片、当前服务商/模型、KI、项目阶段、准确可见错误，以及相关项目事件或日志。删除 API 密钥和私有数据。截图很有帮助，但底层文字和时间戳通常更适合诊断。

GeoForge

把科学知识变成可运行、可检查、可复现的工作流。

更新于 2026 年 8 月 27 日